

Sri Lanka Institute of Information Technology

B.Sc. (Hons) in Information Technology Specialized in
Cyber Security

Systems and Network Programming



Individual Assignment

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Contents

1. Basics of Linux Environments: -----	3
1.1. Virtual Machine Setup-----	3
1.2. Command Line Introduction -----	9
1.3. System Information and User Management -----	15
2. DHCP, DNS and NTP Services-----	16
2.1 DHCP (Dynamic Host Configuration Protocol)-----	16
2.2 DNS (Domain Name System)-----	25
2.3 NTP (Network Time Protocol)-----	34
2.4 Explanation-----	39
3. Security and other servers-----	41
3.1 Shell Scripting-----	43
3.2 SSH (Secure Shell)-----	54
3.3 iptables and ACLs-----	56
3.4 Web Server (Apache/Nginx)-----	62
3.5 Email Server (postfix/Exim)-----	66
4. Linux GDB-----	71
4.a Execution Process-----	71
4.b Debugging Process-----	74
4.c File System Analysis-----	80
4.d Analysis of "data.txt"-----	84

1. Basics of Linux Environments:

1.1. Virtual Machine Setup

Install Virtual Machine Software.

Firstly, we need to install virtualization software that will allow us to run the virtual operating system on our current computer. For this, we can choose one of the two popular choices:

1. Oracle VirtualBox: - Free open-source software for beginners.



(<https://www.virtualbox.org/wiki/Downloads>)

Or

2. VMware Workstation Player: Free version of VMware for personal non-commercial use.



(<https://www.vmware.com/products/workstation-player/workstation-player-evaluation.html>)

When I chose to install Oracle VirtualBox,

Installation steps:

1. To download the software, go to the official website.

(VirtualBox: <https://www.virtualbox.org/wiki/Downloads>)

2. Then run the installer.

-Locate the downloaded file and double-click it to open the installer.

-To install the program, first execute the setup wizard. Then click through the steps using the pre-existing setting by selecting Next and Install.

-Once the installation is complete, open the application.

3. Finally, when you open VirtualBox to confirm successful installation

Download Linux ISO File

Secondly, we have to download the Linux operating system image in **[.iso]** format. This file is actually a virtual CD/DVD that is used to install the operating system inside the VM. When we choose a Linux distribution, we can use

1. Ubuntu (Recommended for first timers):

(<https://ubuntu.com/download/desktop>)



Or

2. CentOS (Community-built version of Red Hat Linux):

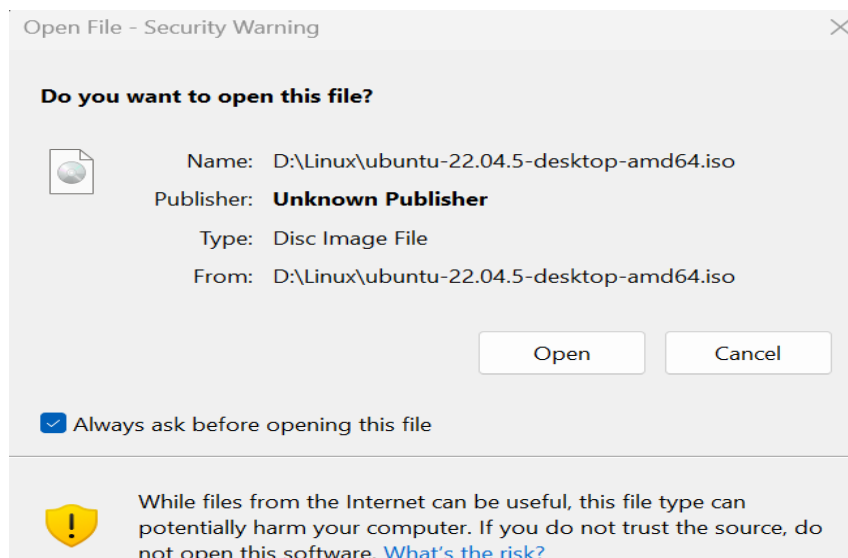
(<https://www.centos.org/download/>)



When I chose Ubuntu,

Download Steps:

1. Visit the official page of Ubuntu.
2. Then click on the download link of the desktop version of the **[.iso]** file.
3. Save the ISO file to a place where you can easily access it.

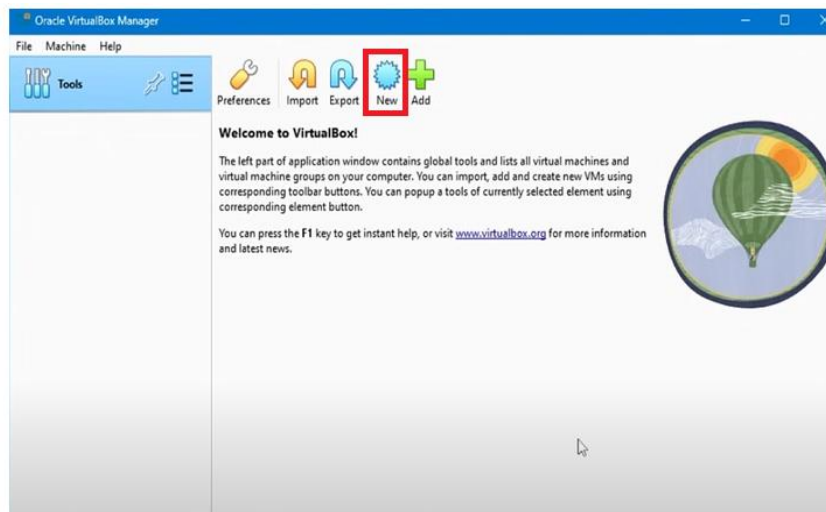


Create a New Virtual Machine

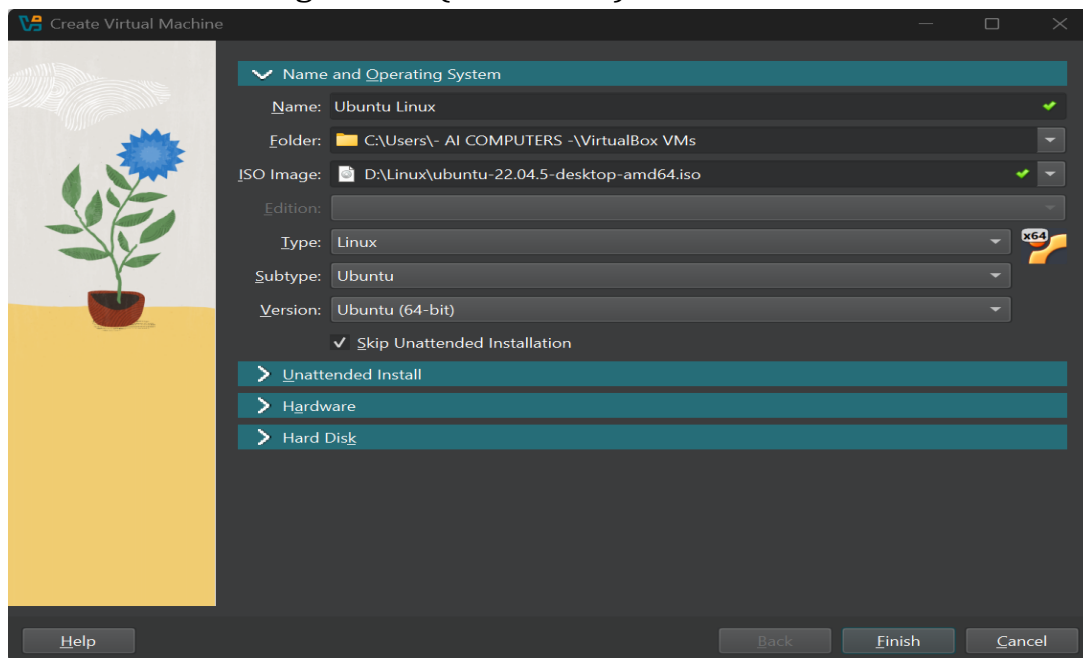
Now that the ISO file is ready, the next step is to create a new virtual machine where the Linux OS will be installed.

Steps to create the VM:

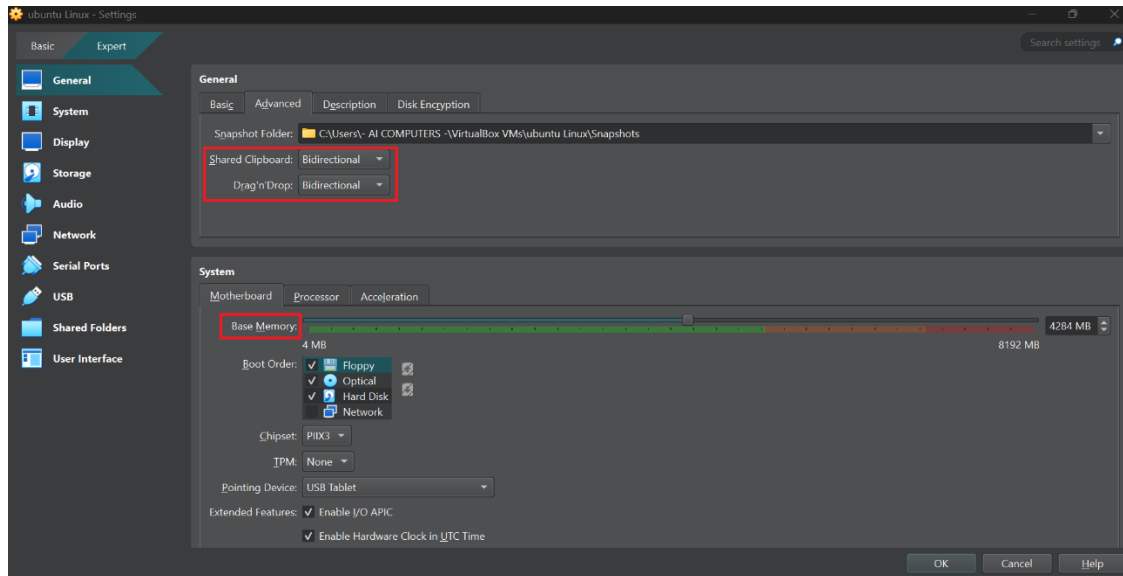
1. Open VirtualBox.
2. Now, click the “New” button to create a new virtual machine.



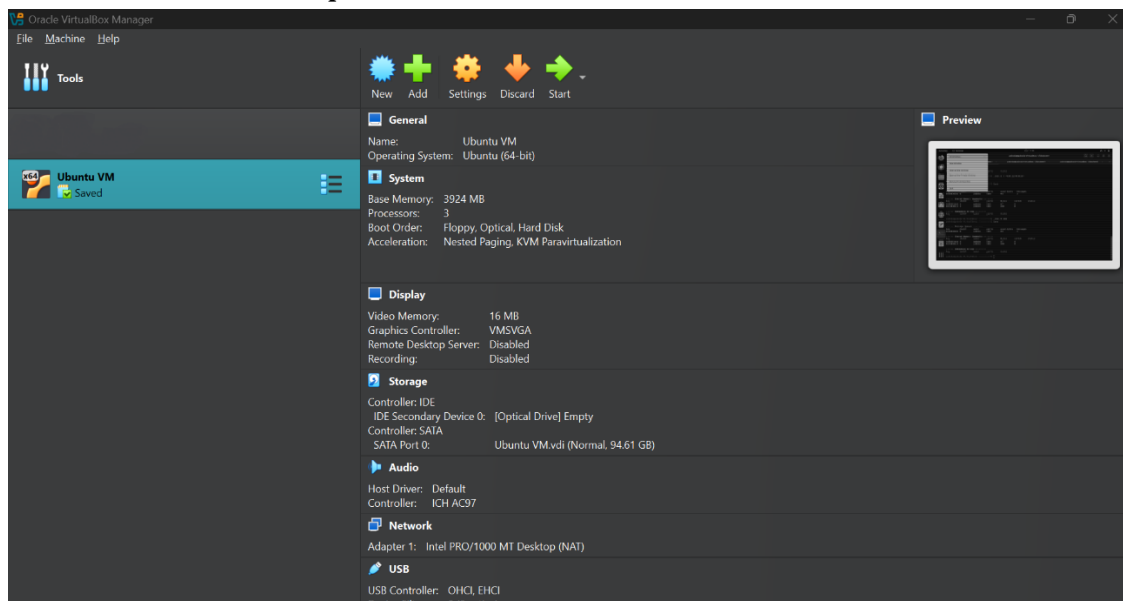
3. Enter the following details (like below):



4. Finally, allocate system resources (like below):



5. During the time of setup or after creating the VM, use the downloaded ISO file as the startup disk.

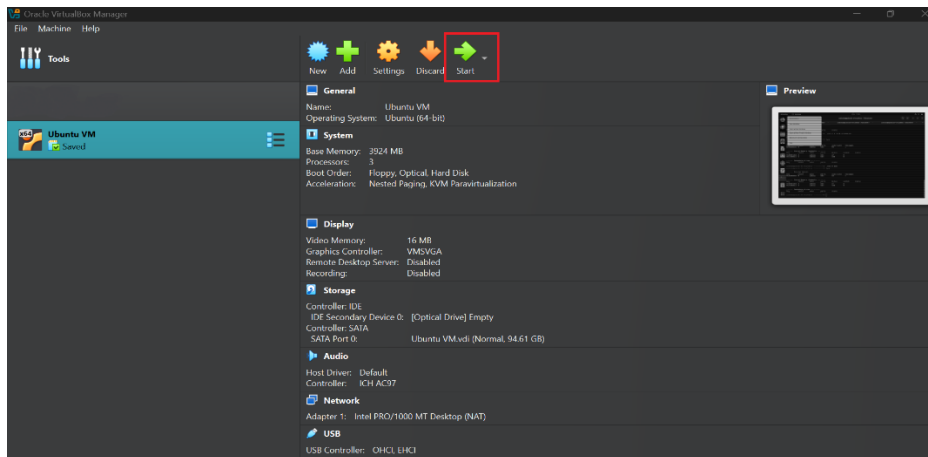


Install the Linux Operating System.

Now, with everything set up, we begin the Linux OS installation process.

Steps:

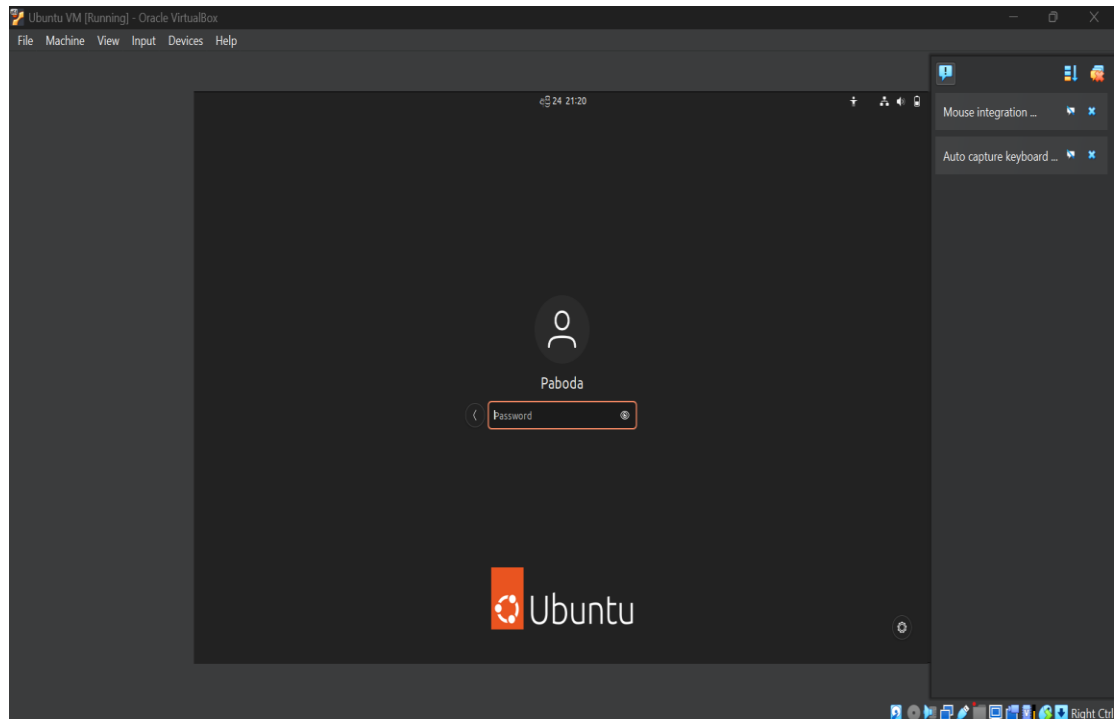
1. Start the VM by clicking start.



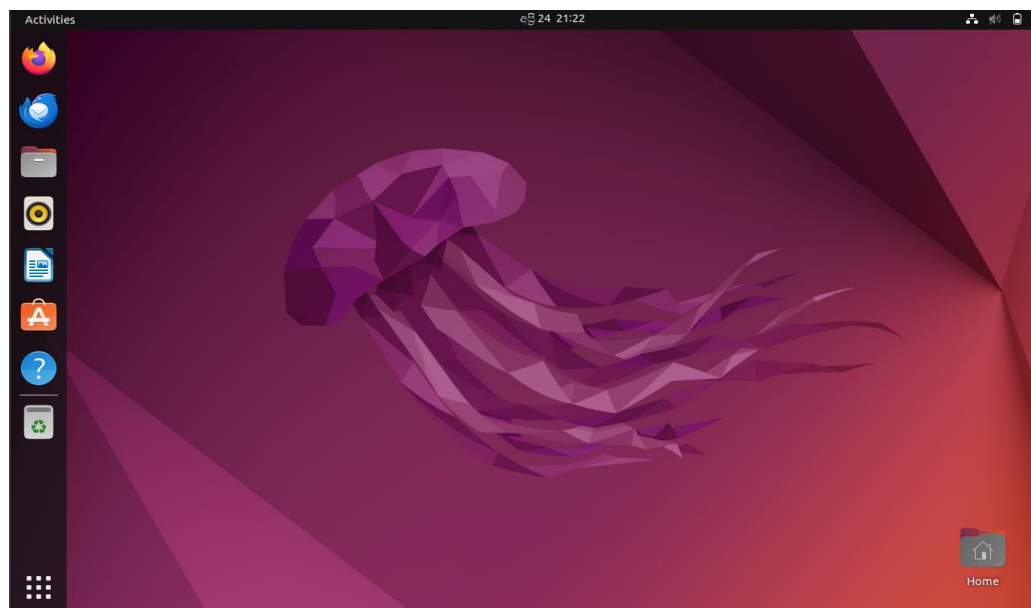
2. The VM will boot up using the ISO file and load the Linux installer.

3. Follow the installation wizard:

- Select Language: e.g. English.
- Click install Ubuntu.
- Provide the required user details:
 - Your Name.
 - Computer Name.
 - Username: e.g. Student.
 - Password: Enter a good password and do not forget it.
- Continue the installation:
 - Wait for files to be copied and installed, which may take a few Minutes. After that, the VM will prompt you to restart. Hereafter Restarting the Linux desktop will boot, and you can now use your new virtual Linux system.



(You have to enter your given password like above to log in.)



(After the logging desktop screen like above)

1.2 Command line Introduction.

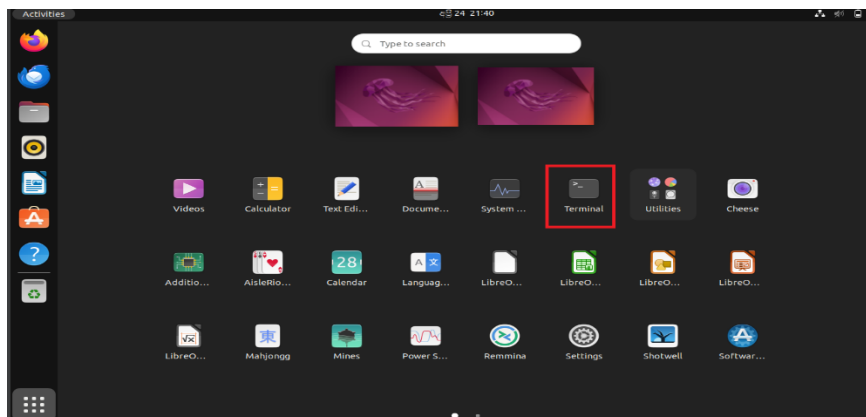
Basic Linux Commands

Following a successful installation and login to the Linux operating system, the second step is familiarization with the command-line interface (CLI), also referred to as the Terminal. In Linux, a lot of strong activity can be achieved through terminal commands.

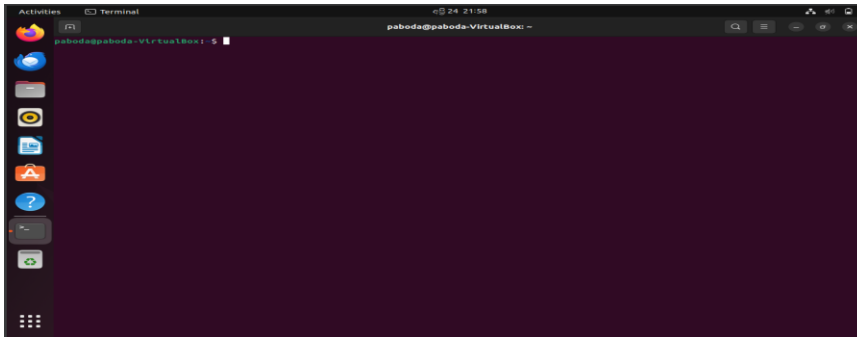
This section addresses 15 basic and essential Linux commands that help navigate the file system, create/modify files, and handle users. These are elementary to understand for beginners to comprehend how Linux works.

Steps to opening the terminal:

1. On the desktop in the bottom of left corner, we can see an icon. So, click once and you can open it like below.



After that, double-click on the terminal icon. Then you can open the terminal like below,



For each command listed below:

- Type the command in the terminal.
- Press Enter to execute.

List of Basic Linux Commands

1. “whoami” - Get the active username.

```
paboda@paboda-VirtualBox:~$ whoami  
paboda
```

2. “who” - Information about users who are currently logged in.

```
paboda@paboda-VirtualBox:~$ who  
paboda    tty2          2025-04-24 22:43 (tty2)
```

3. “pwd” – Print current working directory.

```
paboda@paboda-VirtualBox:~$ pwd  
/home/paboda
```

4. “man” – look for the manual of the command (man who)

```
paboda@paboda-VirtualBox:~$ man  
What manual page do you want?  
For example, try 'man man'.
```

5. “mkdir” – create a directory/a folder (mkdir testfolder)
6. “cd” – change the working directory.

- “cd ..” – move backwards (step by step).

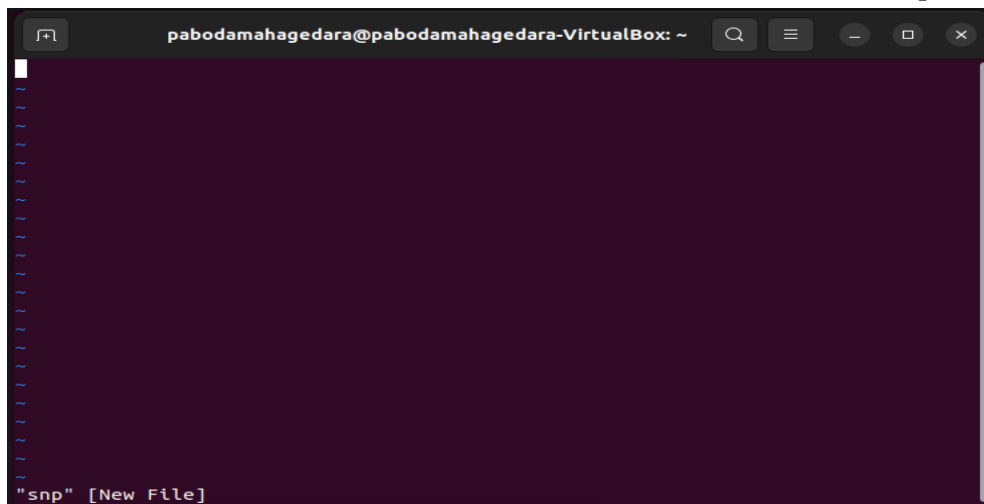
7. “rmdir” – remove directory with all contents (rmdir testfolder)

```
paboda@paboda-VirtualBox:~$ mkdir snpa
paboda@paboda-VirtualBox:~$ cd snpa
paboda@paboda-VirtualBox:~/snpa$ cd ..
paboda@paboda-VirtualBox:~$ rmdir snpa
paboda@paboda-VirtualBox:~$ cd snpa
bash: cd: snpa: No such file or directory
paboda@paboda-VirtualBox:~$
```

8. “vi” – open terminal text editor (vi test file or vi testfile.txt).

Press “i”. After that, you can type anything in there. To save that file backwards, give the command “Press Esc and then “:wq” and again press Enter”. When you provide this command, you can return to the normal terminal by saving that file.

```
paboda@paboda-VirtualBox:~$ vi snp -Input.
```

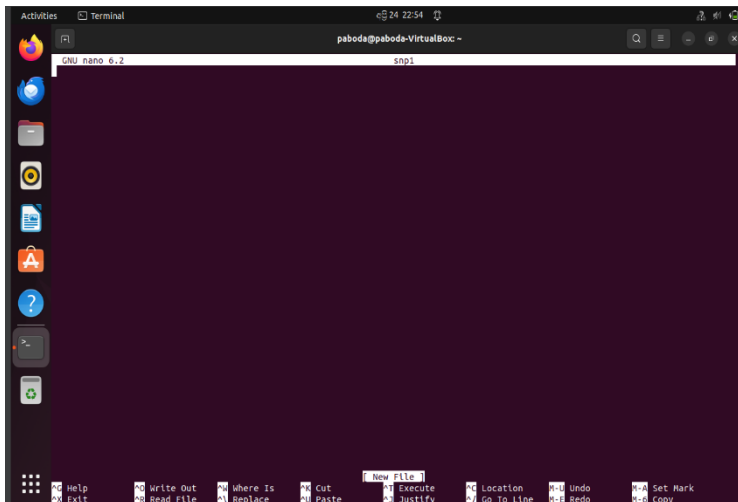


-Output

9. “nano” – another terminal text editor (nano test file or nano testfile.txt).

```
paboda@paboda-VirtualBox:~$ nano snp1
```

-Input



-Output

10. “rm” – remove file (rm test file)

```
rm
paboda@paboda-VirtualBox:~$ ls
Desktop  Downloads  lab5  labsheet7  oddeven.sh  posneg.sh  snap  Templates  Videos
Documents  lab4      lab7  Music      Pictures     Public     snp    usrinfo.sh
paboda@paboda-VirtualBox:~$ rm snp
paboda@paboda-VirtualBox:~$ ls
Desktop  Downloads  lab5  labsheet7  oddeven.sh  posneg.sh  snap    usrinfo.sh
Documents  lab4      lab7  Music      Pictures     Public     Templates  Videos
```

11. “cat” – chow contents of the file (cat test file)

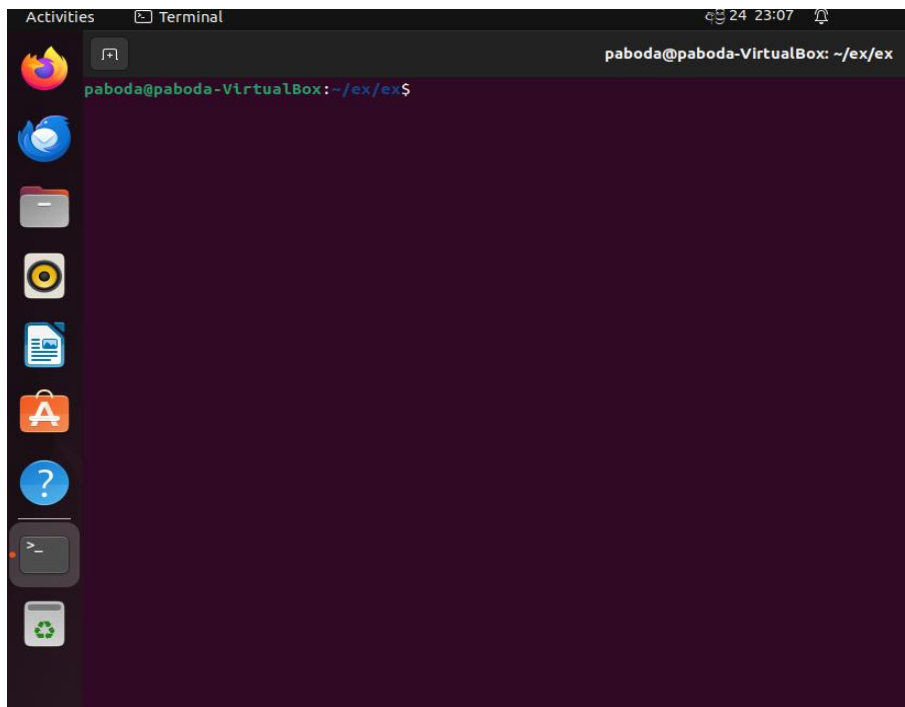
```
paboda@paboda-VirtualBox:~$ mkdir ex
paboda@paboda-VirtualBox:~$ cd ex
paboda@paboda-VirtualBox:~/ex$ cat >ex.txt
A b c          d
```


12. “touch” – create an empty file (touch test file)

13. “clear” – clear the terminal window.

```
paboda@paboda-VirtualBox:~/ex$ mkdir ex
paboda@paboda-VirtualBox:~/ex$ cd ex
paboda@paboda-VirtualBox:~/ex/ex$ touch ex1.txt
paboda@paboda-VirtualBox:~/ex/ex$ rm ex1.txt
paboda@paboda-VirtualBox:~/ex/ex$ clear
```

-Input



- Output.

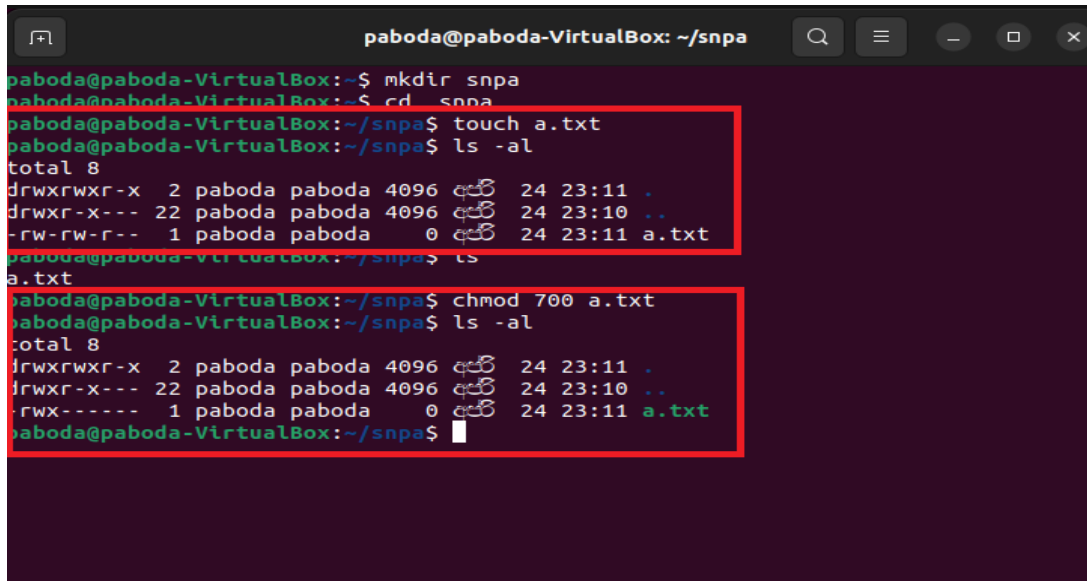
14. “chmod”- grand RWX permission (chmod +X testfile)

In there r, w, x means,

r- read (4)

w- write (2)

x- execute (1)



```
paboda@paboda-VirtualBox: ~/snpa
paboda@paboda-VirtualBox:~$ mkdir snpa
paboda@paboda-VirtualBox:~$ cd snpa
paboda@paboda-VirtualBox:~/snpa$ touch a.txt
paboda@paboda-VirtualBox:~/snpa$ ls -al
total 8
drwxrwxr-x  2 paboda paboda 4096  24 23:11 .
drwxr-x--- 22 paboda paboda 4096  24 23:10 ..
-rw-rw-r--  1 paboda paboda    0  24 23:11 a.txt
paboda@paboda-VirtualBox:~/snpa$ ls
a.txt
paboda@paboda-VirtualBox:~/snpa$ chmod 700 a.txt
paboda@paboda-VirtualBox:~/snpa$ ls -al
total 8
drwxrwxr-x  2 paboda paboda 4096  24 23:11 .
drwxr-x--- 22 paboda paboda 4096  24 23:10 ..
-rwx-----  1 paboda paboda    0  24 23:11 a.txt
paboda@paboda-VirtualBox:~/snpa$
```

If you want to change any permissions, below are the most popular utilized numbers,

- **“700”** – Only the owner can access.
- **“755”** – Owner can w, r and x, but others can only r and x.
- **“777”** – Anyone can r, w and x.

1.3 System Information and User Management

15. "ls" – list files and directories

In there,

Blue color represents- Directories.

White color represents– Files.

Green color represents – Hidden files.

```
paboda@paboda-VirtualBox:~$ ls
Desktop  lab4  labsheet7  Pictures  snap  Videos
Documents lab5  Music      posneg.sh  Templates
Downloads lab7  oddeven.sh Public      usrinfo.sh
```

- "ls -al" – list all files and directories with size, permission and date.

```
paboda@paboda-VirtualBox:~$ ls -al
total 112
drwxr-xr-x 18 paboda paboda 4096 26 16:00 .
drwxr-xr-x  3 root   root   4096 17 15:14 ..
-rw-r--r--  1 paboda paboda   606 26 15:58 .bash_history
-rw-r--r--  1 paboda paboda   220 17 15:14 .bash_logout
-rw-r--r--  1 paboda paboda  3771 17 15:14 .bashrc
drwx----- 11 paboda paboda 4096 17 16:28 .cache
drwx----- 13 paboda paboda 4096 26 15:55 .config
drwxr-xr-x  2 paboda paboda 4096 17 16:27 Desktop
drwxr-xr-x  2 paboda paboda 4096 17 16:27 Documents
drwxr-xr-x  2 paboda paboda 4096 26 15:53 Downloads
drwxr-xr-x  2 paboda paboda 4096 26 15:02 lab4
drwxr-xr-x  2 paboda paboda 4096 26 12:21 lab5
drwxr-xr-x  2 paboda paboda 4096 26 15:58 lab7
drwxr-xr-x  2 paboda paboda 4096  2 11:28 labsheet7
-rw-r--r--  1 paboda paboda   20 20 08:50 .lessht
drwx-----  3 paboda paboda 4096 17 16:27 .local
drwxr-xr-x  2 paboda paboda 4096 17 16:27 Music
-rwxr-xr-x  1 paboda paboda   224 20 10:40 oddeven.sh
drwxr-xr-x  2 paboda paboda 4096 17 16:27 Pictures
-rw-rw-r--  1 paboda paboda   216 20 10:50 posneg.sh
-rw-r--r--  1 paboda paboda   807 17 15:14 .profile
drwxr-xr-x  2 paboda paboda 4096 17 16:27 Public
drwx-----  4 paboda paboda 4096 26 11:12 snap
-rw-r--r--  1 paboda paboda    0 20 09:50 .sudo_as_admin_successful
drwxr-xr-x  2 paboda paboda 4096 17 16:27 Templates
-rwx-----  1 paboda paboda   170 20 09:59 usrinfo.sh
drwxr-xr-x  2 paboda paboda 4096 17 16:27 Videos
-rw-r--r--  1 paboda paboda  7482 20 10:50 .viminfo
```

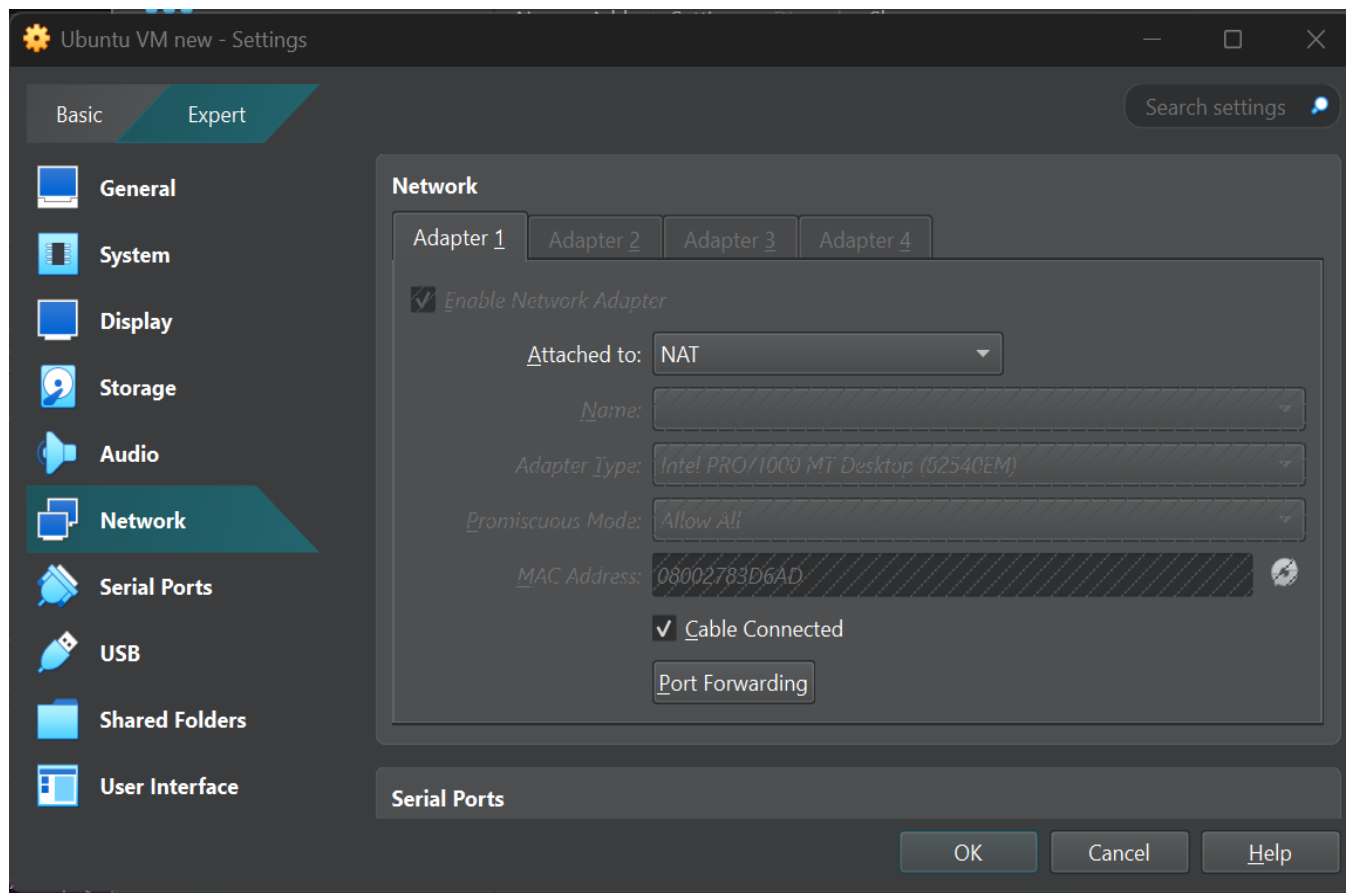
However, these commands are helpful to understand system navigation, file creation and manipulation and basic user management. As well as it is especially important to effectively interact with Linux environments.

2. DHCP, DNS and NTP Services

2.1. DHCP (Dynamic Host Configuration Protocol)

DHCP server installation steps,

- First step is checking the virtual box (Network settings) adapter is NAT.



- Update your system

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://lk.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
21 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-headers-6.8.0-59-generic linux-hwe-6.8-headers-6.8.0-59
  linux-hwe-6.8-tools-6.8.0-59 linux-image-6.8.0-59-generic
  linux-modules-6.8.0-59-generic linux-modules-extra-6.8.0-59-generic
  linux-tools-6.8.0-59-generic
The following packages have been kept back:
  grub-efi-amd64-bin grub-efi-amd64-signed
The following packages will be upgraded:
  distro-info-data gnome-shell gnome-shell-common libnautilus-extension1a
  libpoppler-cpp0v5 libpoppler-glib8 libpoppler118 libxml2
  linux-generic-hwe-22.04 linux-headers-generic-hwe-22.04
  linux-image-generic-hwe-22.04 linux-tools-common nautilus nautilus-data
  poppler-utils ubuntu-advantage-tools ubuntu-pro-client
  ubuntu-pro-client-l10n xwayland
19 upgraded, 7 newly installed, 0 to remove and 2 not upgraded.
5 standard LTS security updates
Need to get 148 MB of archives.
After this operation, 766 MB of additional disk space will be used.
Get:1 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 distro-info-data all 0.52ubuntu0.9 [5,336 B]
Get:2 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libxml2 amd64 2.9.13+dfsg-1ubuntu0.7 [763 kB]
Get:3 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ubuntu-pro-client-l10n amd64 35.1ubuntu0-22.04 [20.6 kB]
Get:4 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ubuntu-pro-client amd64 35.1ubuntu0-22.04 [236 kB]
Get:5 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ubuntu-advantage-tools all 35.1ubuntu0-22.04 [10.9 kB]
Get:6 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gnome-shell amd64 42.9-0ubuntu2.3 [877 kB]
Get:7 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gnome-shell-common all 42.9-0ubuntu2.3 [183 kB]
Get:8 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 nautilus amd64 1:42.6-0ubuntu2 [632 kB]
Get:9 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 nautilus-data all 1:42.6-0ubuntu2 [11.4 kB]
```

- Install the DHCP Server Package.

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install isc-dhcp-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
isc-dhcp-server is already the newest version (4.4.1-2.3ubuntu2.4).
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40
  linux-hwe-6.8-tools-6.8.0-40 linux-image-6.8.0-40-generic
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic
  linux-tools-6.8.0-40-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

- Identify your Network Interface

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:83:d6:ad brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85934sec preferred_lft 85934sec
    inet6 fd00::9289:757d:47a1:dc5e/64 scope global temporary dynamic
        valid_lft 85936sec preferred_lft 13936sec
    inet6 fd00::a00:27ff:fe83:d6ad/64 scope global dynamic mngtmpaddr
        valid_lft 85936sec preferred_lft 13936sec
    inet6 fe80::a00:27ff:fe83:d6ad/64 scope link
        valid_lft forever preferred_lft forever
```

- Configure DHCP Server

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/dhcp/dhcpd.conf
```

- Add or update this configuration.(customized to match your 10.0.2.0/24):

```
GNU nano 6.2 /etc/dhcp/dhcpd.conf
#
#   members of "foo";
#   range 10.17.224.10 10.17.224.250;
#   }
#   pool {
#       deny members of "foo";
#       range 10.0.29.10 10.0.29.230;
#   }
#}

subnet 10.0.2.0 netmask 255.255.255.0 {
    range 10.0.2.100 10.0.2.200;
    option routers 10.0.2.1;
    option subnet-mask 255.255.255.0;
    option domain-name-servers 8.8.8.8, 8.8.4.4;
}

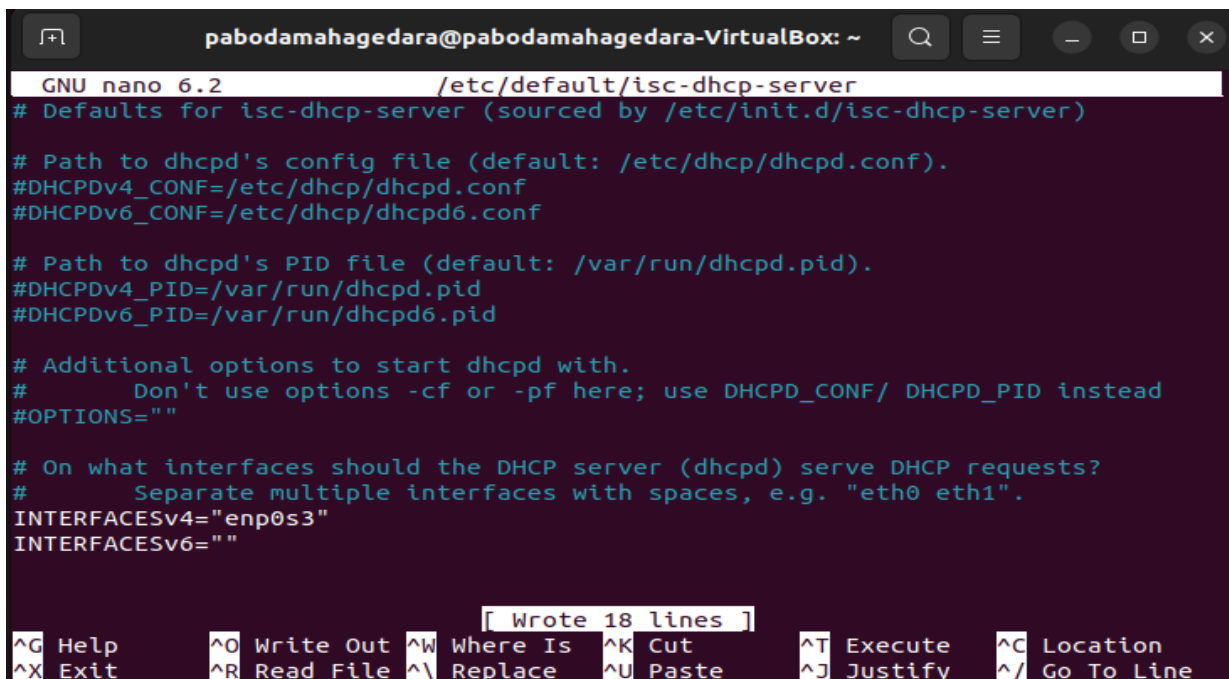
[ Wrote 121 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

- Set DHCP to use your interface

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/default/isc-dhcp-server
```

-to specify the interface (enp0s3) the DHCP server will listen on.

-Find the **INTERFACESv4** line and update with your network interface **enp0s3**



```
GNU nano 6.2 /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

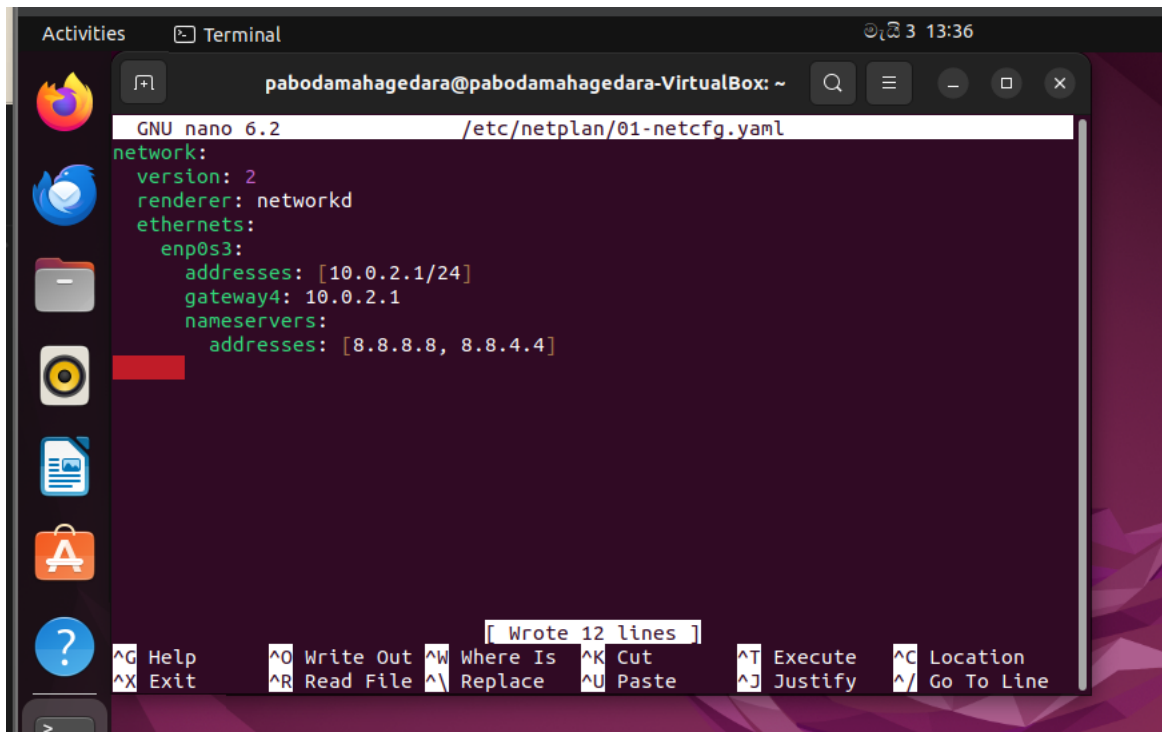
# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s3"
INTERFACESv6=""

[ Wrote 18 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

- Assign a Static IP to server.

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~$ sudo nano /etc/netplan/01-netcfg.yaml
```

-Update like below;



```
GNU nano 6.2 /etc/netplan/01-netcfg.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      addresses: [10.0.2.1/24]
      gateway4: 10.0.2.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

-Apply the changes;


```
pabodamahagedara@pabodamahagedara-VirtualBox: ~
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo netplan apply
** (generate:12028): WARNING **: 13:37:21.558: Permissions for /etc/netplan/00-installer-confi
tion should NOT be accessible by others.
** (generate:12028): WARNING **: 13:37:21.559: Permissions for /etc/netplan/01-netcfg.yaml are
d NOT be accessible by others.
** (generate:12028): WARNING **: 13:37:21.559: `gateway4` has been deprecated, use default rou
See the 'Default routes' section of the documentation for more details.
** (generate:12028): WARNING **: 13:37:21.559: Permissions for /etc/netplan/01-network-manager
uration should NOT be accessible by others.
** (process:12026): WARNING **: 13:37:22.145: Permissions for /etc/netplan/00-installer-confi
ion should NOT be accessible by others.
** (process:12026): WARNING **: 13:37:22.146: Permissions for /etc/netplan/01-netcfg.yaml are
NOT be accessible by others.
** (process:12026): WARNING **: 13:37:22.146: `gateway4` has been deprecated, use default rou
See the 'Default routes' section of the documentation for more details.
** (process:12026): WARNING **: 13:37:22.146: Permissions for /etc/netplan/01-network-manager-
ration should NOT be accessible by others.
** (process:12026): WARNING **: 13:37:22.408: Permissions for /etc/netplan/00-installer-confi
ion should NOT be accessible by others.
** (process:12026): WARNING **: 13:37:22.409: Permissions for /etc/netplan/01-netcfg.yaml are
NOT be accessible by others.
** (process:12026): WARNING **: 13:37:22.409: `gateway4` has been deprecated, use default rou
See the 'Default routes' section of the documentation for more details.
** (process:12026): WARNING **: 13:37:22.409: Permissions for /etc/netplan/01-network-manager-
ration should NOT be accessible by others.
** (process:12026): WARNING **: 13:37:22.409: Permissions for /etc/netplan/00-installer-confi
ion should NOT be accessible by others.
```

- Restart and Enable DHCP Server
 - Restart the DHCP server to apply the new configuration

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl restart isc-dhcp-server
```

- Enable the DHCP server to start at boot

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable isc-dhcp-server
```

- Verify the DHCP Server
 - Check the status of the DHCP server to ensure it's running;

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl status isc-dhcp-se
●isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; vend
   Active: active (running) since Sat 2025-05-03 13:37:57 +0530; 1min 22s ago
     Docs: man:dhcpd(8)
    Main PID: 12152 (dhcpd)
      Tasks: 4 (limit: 2269)
     Memory: 6.1M
        CPU: 14ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─12152 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/>

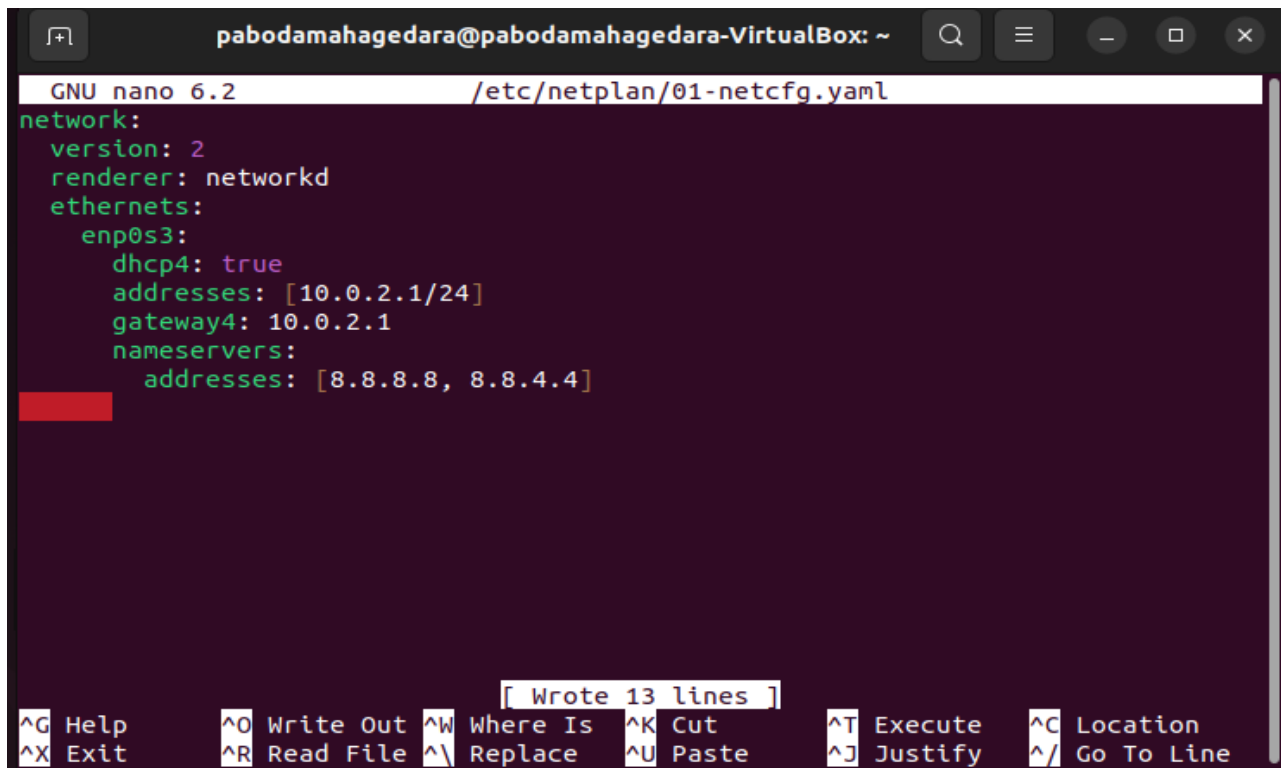
03 13:37:57 pabodamahagedara-VirtualBox sh[12152]: PID file: /run/dhcp-serv>
03 13:37:57 pabodamahagedara-VirtualBox dhcpd[12152]: Wrote 0 leases to lea>
03 13:37:57 pabodamahagedara-VirtualBox sh[12152]: Wrote 0 leases to leases>
03 13:37:57 pabodamahagedara-VirtualBox dhcpd[12152]: Listening on LPF/enp0>
03 13:37:57 pabodamahagedara-VirtualBox sh[12152]: Listening on LPF/enp0s3/>
03 13:37:57 pabodamahagedara-VirtualBox sh[12152]: Sending on LPF/enp0s3/>
03 13:37:57 pabodamahagedara-VirtualBox sh[12152]: Sending on Socket/fall>
03 13:37:57 pabodamahagedara-VirtualBox dhcpd[12152]: Sending on LPF/enp0>
03 13:37:57 pabodamahagedara-VirtualBox dhcpd[12152]: Sending on Socket/f>
03 13:37:57 pabodamahagedara-VirtualBox dhcpd[12152]: Server starting servi>
lines 1-21/21 (END)
[1]+  Stopped                  sudo systemctl status isc-dhcp-server
pabodamahagedara@pabodamahagedara-VirtualBox:~$
```

DHCP Client Machine Configuration

- Open Netplan config:

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/netplan/01-netcfg
.yaml
[sudo] password for pabodamahagedara:
```

- Set DHCP for interface:



The screenshot shows a terminal window titled "pabodamahagedara@pabodamahagedara-VirtualBox: ~". Inside the terminal, the GNU nano 6.2 editor is open, editing the file "/etc/netplan/01-netcfg.yaml". The configuration is as follows:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: true
      addresses: [10.0.2.1/24]
      gateway4: 10.0.2.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

A red cursor is visible on the line containing "addresses: [8.8.8.8, 8.8.4.4]". At the bottom of the terminal, a status bar indicates "[Wrote 13 lines]". Below the status bar, a list of keyboard shortcuts is displayed:

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

- Save & Apply:

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo netplan apply

** (generate:1969): WARNING **: 21:38:42.619: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible by others.

** (generate:1969): WARNING **: 21:38:42.619: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.

** (generate:1969): WARNING **: 21:38:42.620: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (generate:1969): WARNING **: 21:38:42.620: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.

** (process:1967): WARNING **: 21:38:43.154: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible by others.

** (process:1967): WARNING **: 21:38:43.155: Permissions for /etc/netplan/01-net
```

- Check if IP is assigned

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:83:d6:ad brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.1/24 brd 10.0.2.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet 10.0.2.15/24 brd 10.0.2.255 scope global secondary dynamic noprefixroute enp0s3
        valid_lft 86390sec preferred_lft 86390sec
    inet6 fd00::d545:503b:3ce2:7cca/64 scope global temporary dynamic
        valid_lft 86392sec preferred_lft 14392sec
    inet6 fd00::a00:27ff:fe83:d6ad/64 scope global dynamic mngtmpaddr
        valid_lft 86392sec preferred_lft 14392sec
    inet6 fe80::a00:27ff:fe83:d6ad/64 scope link
        valid_lft forever preferred_lft forever
```

- We can see an IP in the range assigned by this DHCP server.

2.2 DNS (Domain Name System)

Use BIND9, which is the most common DNS server for Ubuntu.

- Install BIND9 DNS Server

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~  
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt update  
Hit:1 http://lk.archive.ubuntu.com/ubuntu jammy InRelease  
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]  
Hit:3 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:4 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,306 kB]  
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [632 kB]  
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [351 kB]  
Get:8 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [3,353 kB]  
Get:9 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [600 kB]  
Get:10 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [974 kB]  
Get:11 http://security.ubuntu.com/ubuntu jammy-security/universe i386 Packages [658 kB]  
Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [210 kB]  
Fetched 9,213 kB in 21s (439 kB/s)  
Reading package lists... Done  
Building dependency tree... Done
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install bind9 bind9utils bind9-doc -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
bind9 is already the newest version (1:9.18.30-0ubuntu0.22.04.2).
bind9-doc is already the newest version (1:9.18.30-0ubuntu0.22.04.2).
bind9utils is already the newest version (1:9.18.30-0ubuntu0.22.04.2).
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40
  linux-hwe-6.8-tools-6.8.0-40 linux-image-6.8.0-40-generic
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic
  linux-tools-6.8.0-40-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

- Define a DNS Zone
 - Edit the local zone definition

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/bind/named.conf.local
```

-Add this block within replacing your custom domain.

```
GNU nano 6.2 /etc/bind/named.conf.local
//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "snp.local" {
    type master;
    file "/etc/bind/db.snp.local";
};

[ Wrote 13 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

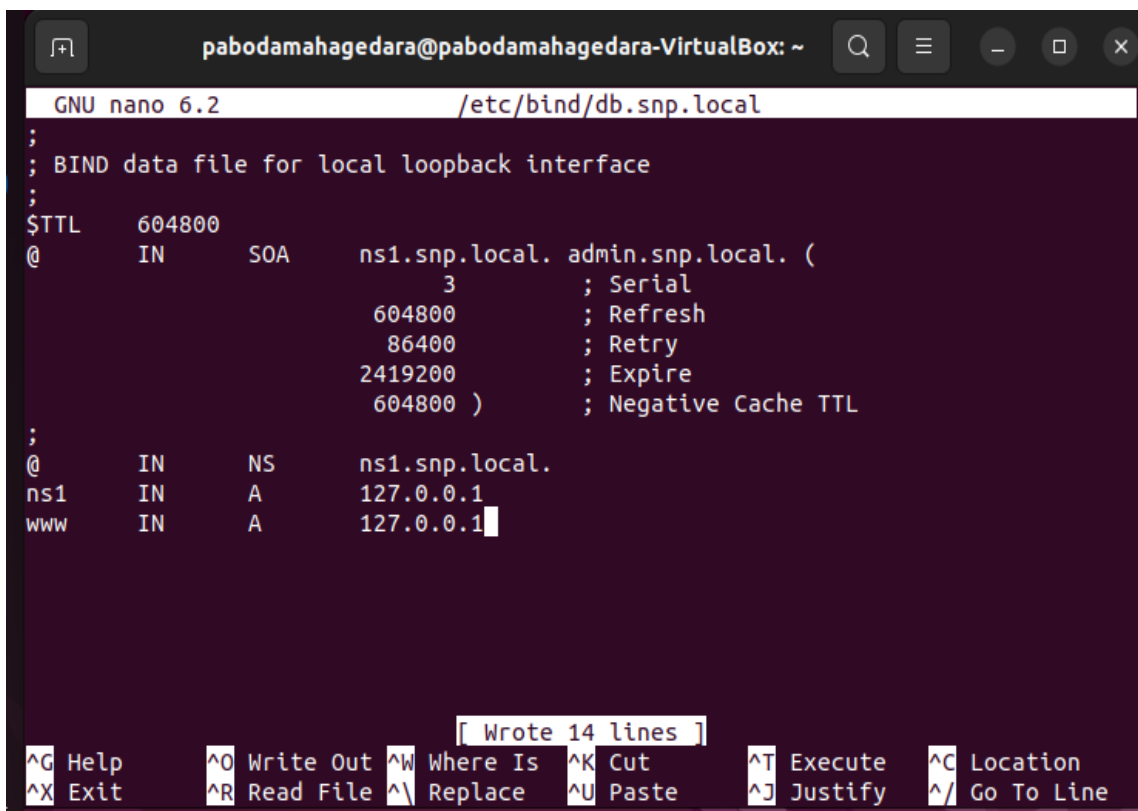

- Create the Zone file.
 - Copy the default template.

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo cp /etc/bind/db.local /etc/  
bind/db.snp.local  
[sudo] password for pabodamahagedara:
```

- Edit the new file.

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/bind/db.snp.local
```

- Replace content like this.



```
GNU nano 6.2 /etc/bind/db.snp.local  
;  
; BIND data file for local loopback interface  
;  
$TTL      604800  
@         IN      SOA      ns1.snp.local. admin.snp.local. (  
          3        ; Serial  
          604800    ; Refresh  
          86400     ; Retry  
          2419200   ; Expire  
          604800 )   ; Negative Cache TTL  
;  
@         IN      NS       ns1.snp.local.  
ns1       IN      A        127.0.0.1  
www       IN      A        127.0.0.1  
[ Wrote 14 lines ]  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location  
^X Exit      ^R Read File  ^\ Replace    ^U Paste       ^J Justify    ^_ Go To Line
```

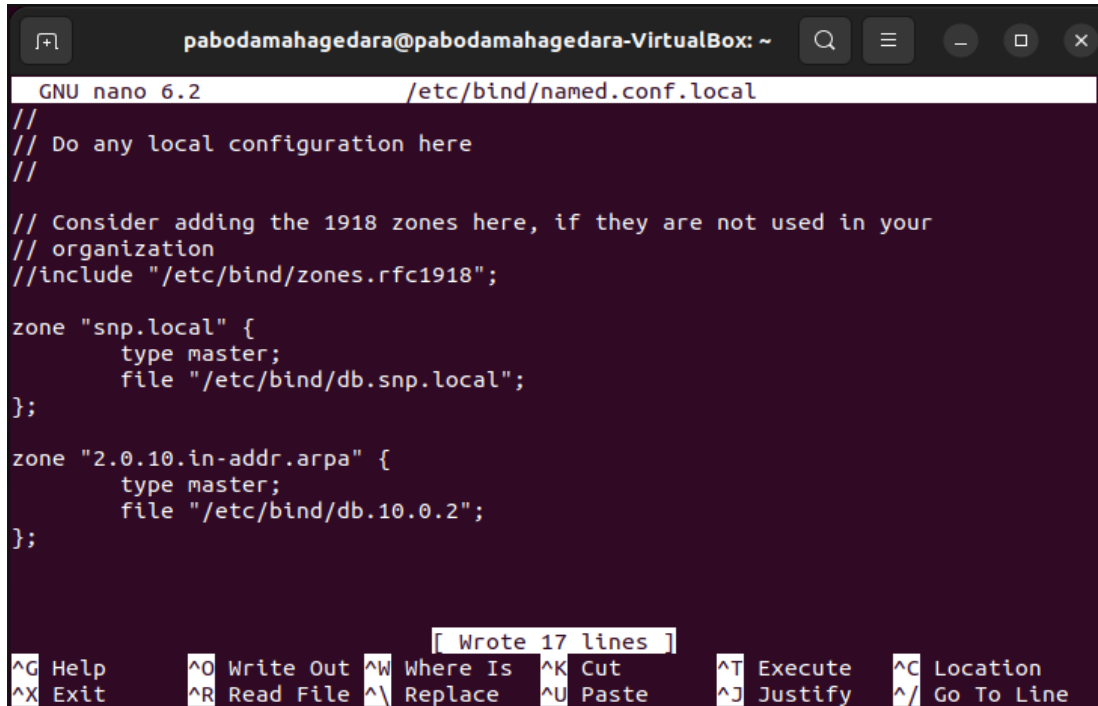
- 127.0.0.1: DNS server IP
- 127.0.0.1: a web or client machine IP

- (Optional) Configure Reverse Zone

-Add reverse zone to **named.conf.local**

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/bind/named.conf.local
```

-Add



```
GNU nano 6.2 /etc/bind/named.conf.local
//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "snp.local" {
    type master;
    file "/etc/bind/db.snp.local";
};

zone "2.0.10.in-addr.arpa" {
    type master;
    file "/etc/bind/db.10.0.2";
};

[ Wrote 17 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

-Create reverse zone file

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo cp /etc/bind/db.127 /etc/bind/db.10.0.2
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/bind/db.10.0.2
```


-Update contents

```

GNU nano 6.2 /etc/bind/db.10.0.2
;
; BIND reverse data file for local loopback interface
;
$TTL      604800
@        IN      SOA      ns1.snp.local. admin.snp.local. (
                        1      ; Serial
                        604800 ; Refresh
                        86400  ; Retry
                        2419200 ; Expire
                        604800 ) ; Negative Cache TTL
;
@        IN      NS       ns1.snp.local.
15       IN      PTR      ns1.snp.local.
20       IN      PTR      www.snp.local.

[ Wrote 14 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line

```

- Check BIND Configuration

```

pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo named-checkconf
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo named-checkzone snp.local /
etc/bind/db.snp.local
zone snp.local/IN: loaded serial 3
OK
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo named-checkzone 2.0.10.in-a
ddr.arpa /etc/bind/db.10.0.2
zone 2.0.10.in-addr.arpa/IN: loaded serial 1
OK

```

-If no errors, go ahead

- Restart and Enable BIND Service

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl restart bind9
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl enable bind9
Failed to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: ena>
   Active: active (running) since Mon 2025-05-05 23:34:46 +0530; 2min 4s ago
     Docs: man:named(8)
   Process: 3136 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUC>
   Main PID: 3137 (named)
    Tasks: 5 (limit: 2269)
   Memory: 5.5M
      CPU: 25ms
   CGroup: /system.slice/named.service
           └─3137 /usr/sbin/named -u bind

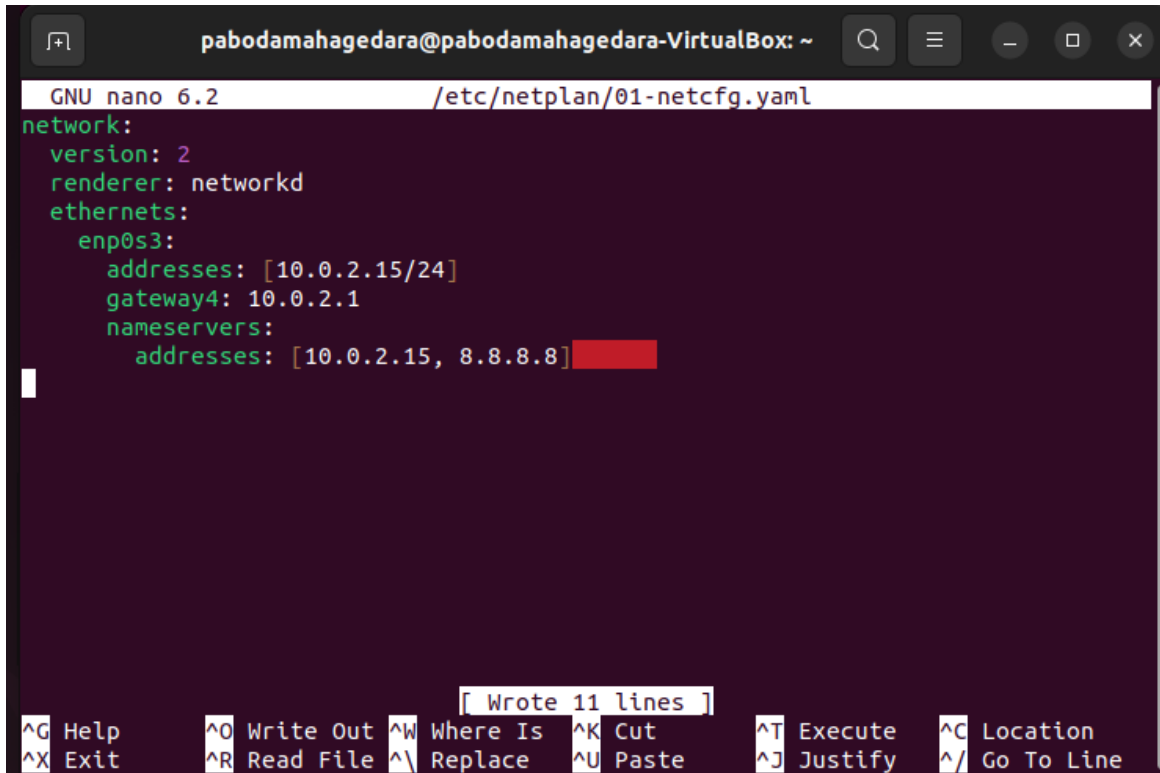
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: zone 255.in-addr.arpa/>
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: zone 2.0.10.in-addr.ar>
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: zone 127.in-addr.arpa/>
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: zone snp.local/IN: loa>
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: zone localhost/IN: loa>
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: all zones loaded
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: running
○ 05 23:34:46 pabodamahagedara-VirtualBox systemd[1]: Started BIND Domain Nam>
○ 05 23:34:46 pabodamahagedara-VirtualBox named[3137]: managed-keys-zone: Key>
```

-Look for: active (running)

- Configure the system to use local DNS

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/netplan/01-netcfg
.yaml
[sudo] password for pabodamahagedara:
```

-set DNS server:



```
GNU nano 6.2 /etc/netplan/01-netcfg.yaml
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      addresses: [10.0.2.15/24]
      gateway4: 10.0.2.1
      nameservers:
        addresses: [10.0.2.15, 8.8.8.8]

```

[Wrote 11 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

-Replace **enp0s3** with your actual interface if different.

-Apply changes,

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo netplan apply

** (generate:3308): WARNING **: 09:06:53.881: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible by others.

** (generate:3308): WARNING **: 09:06:53.882: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.

** (generate:3308): WARNING **: 09:06:53.882: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (generate:3308): WARNING **: 09:06:53.882: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.
```

- Test DNS Resolution

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~
pabodamahagedara@pabodamahagedara-VirtualBox:~$ dig @10.0.2.15 www.snp.local

; <<>> DiG 9.18.30-0ubuntu0.22.04.2-Ubuntu <<>> @10.0.2.15 www.snp.local
; (1 server found)
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 15967
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 82cd9626a0aa45b50100000068198402f35006bef34ee2e8 (good)
;; QUESTION SECTION:
;www.snp.local.                IN      A

;; ANSWER SECTION:
www.snp.local.                604800  IN      A      127.0.0.1

;; Query time: 0 msec
;; SERVER: 10.0.2.15#53(10.0.2.15) (UDP)
;; WHEN: Tue May 06 09:07:38 +0530 2025
;; MSG SIZE rcvd: 86
```



```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ nslookup www.snp.local 10.0.2.15
Server:      10.0.2.15
Address:     10.0.2.15#53

Name:   www.snp.local
Address: 127.0.0.1
```

2.3 NTP (Network Time Protocol)

- Install the NTP package
- Update the system to make sure you have the latest packages

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~  
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt update && sudo apt upgr  
ade -y  
[sudo] password for pabodamahagedara:  
Ign:1 http://lk.archive.ubuntu.com/ubuntu jammy InRelease  
Ign:2 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Ign:3 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Ign:4 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Ign:4 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Ign:1 http://lk.archive.ubuntu.com/ubuntu jammy InRelease  
Ign:2 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Ign:3 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Ign:1 http://lk.archive.ubuntu.com/ubuntu jammy InRelease  
Ign:2 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Ign:3 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Ign:4 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Err:1 http://lk.archive.ubuntu.com/ubuntu jammy InRelease  
  Could not connect to lk.archive.ubuntu.com:80 (185.125.190.83). - connect (113  
: No route to host) Could not connect to lk.archive.ubuntu.com:80 (91.189.91.82)  
 - connect (113: No route to host) Could not connect to lk.archive.ubuntu.com:8  
0 (91.189.91.81). - connect (113: No route to host) Could not connect to lk.arch  
ive.ubuntu.com:80 (91.189.91.83). - connect (113: No route to host) Could not co  
nnect to lk.archive.ubuntu.com:80 (185.125.190.81). - connect (113: No route to  
host) Could not connect to lk.archive.ubuntu.com:80 (185.125.190.82). - connect  
(113: No route to host)
```

- Install the NTP client

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install ntp -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
ntp is already the newest version (1:4.2.8p15+dfsg-1ubuntu2).  
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

- Verify that the **ntp** package is installed

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ dpkg -l | grep ntp  
ii  ntp                                1:4.2.8p15+dfsg-1ubuntu2  
    amd64                          Network Time Protocol daemon and utility programs  
ii  sntp                              1:4.2.8p15+dfsg-1ubuntu2  
    amd64                          Network Time Protocol - sntp client
```

- Start and enable the NTP service
- Start the NTP service

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl start ntp
```

- Enable NTP to start on boot

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl enable ntp
Synchronizing state of ntp.service with SysV service script with /lib/systemd/sy
stemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ntp
```

- Check the service status

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl status ntp
● ntp.service - Network Time Service
   Loaded: loaded (/lib/systemd/system/ntp.service; enabled; vendor preset: ena
   Active: active (running) since Wed 2025-05-07 12:57:33 +0530; 33min ago
     Docs: man:ntpd(8)
    Main PID: 12186 (ntpd)
      Tasks: 2 (limit: 2269)
    Memory: 1.4M
       CPU: 305ms
    CGroup: /system.slice/ntp.service
            └─12186 /usr/sbin/ntpd -p /var/run/ntpd.pid -g -u 131:139

07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: 2620:2d:4000:1::41 loc>
07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: 2001:4860:4806:4:: loc>
07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: 2620:2d:4000:1::3f loc>
07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: 2001:4860:4806:8:: loc>
07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: 2001:4860:4806:c:: loc>
07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: 2001:4860:4806:: local>
07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: Deleting interface #10>
07 13:19:24 pabodamahagedara-VirtualBox ntpd[12186]: Deleting interface #11>
07 13:19:30 pabodamahagedara-VirtualBox ntpd[12186]: Listen normally on 12 >
07 13:19:30 pabodamahagedara-VirtualBox ntpd[12186]: new interface(s) found>
lines 1-21/21 (END)
```


- Check NTP synchronization status
- Verify time synchronization

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ timedatectl status
Local time: 2025-05-07 12:14:18 +0530
Universal time: 2025-05-07 06:44:18 UTC
RTC time: 2025-05-07 06:44:18
Time zone: Asia/Colombo (+0530, +0530)
System clock synchronized: yes
NTP service: n/a
RTC in local TZ: no
```

-Look for **NTP synchronized: yes**

- Check NTP peers (servers)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ntpq -p
```

remote	refid	st	t	when	poll	reach	delay	offset	jitter
0.ubuntu.pool.n	.POOL.	16	p	-	64	0	0.000	+0.000	0.000
1.ubuntu.pool.n	.POOL.	16	p	-	64	0	0.000	+0.000	0.000
2.ubuntu.pool.n	.POOL.	16	p	-	64	0	0.000	+0.000	0.000
3.ubuntu.pool.n	.POOL.	16	p	-	64	0	0.000	+0.000	0.000
ntp.ubuntu.com	.POOL.	16	p	-	64	0	0.000	+0.000	0.000
time.google.com	.POOL.	16	p	-	64	0	0.000	+0.000	0.000

✓ Configure NTP servers (optional)

-If you need to change the default NTP servers:

- Edit the NTP configuration file

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano .etc/ntp.conf
```


- Add

```

pabodamahagedara@pabodamahagedara-VirtualBox: ~
GNU nano 6.2 /etc/ntp.conf
filegen loopstats file loopstats type day enable
filegen peerstats file peerstats type day enable
filegen clockstats file clockstats type day enable

# Specify one or more NTP servers.

# Use servers from the NTP Pool Project. Approved by Ubuntu Technical Board
# on 2011-02-08 (LP: #104525). See http://www.pool.ntp.org/join.html for
# more information.
pool 0.ubuntu.pool.ntp.org iburst
pool 1.ubuntu.pool.ntp.org iburst
pool 2.ubuntu.pool.ntp.org iburst
pool 3.ubuntu.pool.ntp.org iburst

# Use Ubuntu's ntp server as a fallback.
server ntp.ubuntu.com
server time.google.com
# Access control configuration; see /usr/share/doc/ntp-doc/html/acopt.html for
# details. The web page <http://support.ntp.org/bin/view/Support/AccessRestrict>
# might also be helpful.

[ Wrote 60 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

```

- ✓ Restart NTP Service
- Restart NTP service to apply the changes

```

pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl restart ntp
[sudo] password for pabodamahagedara:

```

- ✓ Check NTP synchronization again
-

- After the restarting the service, check if synchronized is now active

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ timedatectl status
Local time: ㉿ 2025-05-07 13:33:11 +0530
Universal time: ㉿ 2025-05-07 08:03:11 UTC
RTC time: ㉿ 2025-05-07 08:03:11
Time zone: Asia/Colombo (+0530, +0530)
System clock synchronized: yes
NTP service: n/a
RTC in local TZ: no
```

- You can also verify the synchronization by running

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ntpq -p
      remote                refid              st t when poll reach  delay  offset  jitter
=====
0.ubuntu.pool.n .POOL.             16 p   -   64    0    0.000  +0.000  0.000
1.ubuntu.pool.n .POOL.             16 p   -   64    0    0.000  +0.000  0.000
2.ubuntu.pool.n .POOL.             16 p   -   64    0    0.000  +0.000  0.000
3.ubuntu.pool.n .POOL.             16 p   -   64    0    0.000  +0.000  0.000
ntp.ubuntu.com .POOL.             16 p   -   64    0    0.000  +0.000  0.000
time.google.com .POOL.             16 p   -   64    0    0.000  +0.000  0.000
*time.cloudflare 10.4.8.56          3 u   23   64   17   38.636  -46.364 135.603
+time.cloudflare 10.4.8.56          3 u   27   64   17   53.413  -4.064 297.738
2620:2d:4000:1: 17.253.28.251    2 u   49   64   35  229.590  -21.845 207.841
2620:2d:4000:1: 183.160.133.132    2 u   51   64   37  231.442  -74.969 110.889
2620:2d:4000:1: 201.68.88.106      2 u   54   64   37  225.840  -80.083  87.093
+time2.google.co .GOOG.             1 u   33   64   17  234.864  -69.411 208.672
+time1.google.co .GOOG.             1 u   34   64   17  234.788  -73.819 206.608
+time4.google.co .GOOG.             1 u   65   64    7   98.948  -24.274 193.297
+time3.google.co .GOOG.             1 u   35   64   17  166.971  -19.633 202.463
```

-If synchronization is working, you should see an asterisk (*) next to one of the NTP servers, indication that it is being used as the source of system's time.

2.4 Need and use of the DHCP, DNS, and NTP network services in the administration of network services:

The fundamental networking utilities (DHCP, DNS, NTP) are all extremely important for any modern network's successful administration and use. All three serve vital functions for free communication, reliability, and automation.

- **DHCP (Dynamic Host Configuration Protocol)**

This would automatically configure the IP address and network configuration on the devices. Admins would have to manually configure these things on each device, which is slow and likely to be incorrect.

-Use in administration:

- Reduces manual IP configuration effort.
- Prevents IP conflicts with dynamic allocation.
- Simplifies network management, especially in large networks.

- **DNS (Domain Name System)**

DNS converts human-readable domain names (www.google.com) to machine-readable IP addresses. Otherwise, without DNS, people would have to memorize IP addresses to reach websites / use services.

-Use in administration:

- Enable easy access to network resources through hostnames.
- Supports internal name resolution within organizations.
- Simplifies service configuration and improves user experience.

- **NTP (Network Time Protocol)**

Syncs the time on all of the computers in the network. The right time is important for logging, security (like certificates), file systems and task synchronization.

-Use in administration:

- Ensure consistent timestamps in logs for troubleshooting and auditing.
- Support for the timely maintenance of time-sensitive operations (backups, scheduled tasks).
- Facilitates the proper functioning of security features.

These services automate, simplify and secure network operations, so they're something almost any network administrator should have.

3. Security and other servers

Tasks:

3.1 Shell Scripting:

- i. Find and delete all log files (.log) in the /var/log/custom_logs/ directory that are older than 7 days

- Open terminal by pressing **Ctrl+Alt+T** or open from the application menu.
- Create the Custom logs directory
-if it doesn't exist already, create it

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo mkdir -p /var/log/custom_logs
```

- Add sample **.log** files for testing

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo touch /var/log/custom_logs/test1.log
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo touch /var/log/custom_logs/test2.log
```

-To simulate old files (older than 7 days):

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo touch -d "10 days ago" /var/log/custom_logs/test1.log
```

- Manually verify old files (before Deleting)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ find /var/log/custom_logs/ -name "*.log" -type f -mtime +7 -print  
/var/log/custom_logs/test1.log
```

-you should see **test1.log** in the output.

- Create a Bash script

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ nano delete_old_logs.sh
```

-Paste the followings into the file:

```
GNU nano 6.2 delete_old_logs.sh *
#!/bin/bash
#task1 (i): Delete. log files older than 7 days from /var/log/custom_logs/

#define the directory
LOG_DIR="/var/log/custom_logs"

#find and delete .log files older than 7 days
DELETED_FILES=$(find "$LOG_DIR" -name "*.log" -type f -mtime +7 -print -delete)

#print summary
echo "Deleted the following .log files older than 7 days:"
echo "$DELETED_FILES"

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-
```

- Save and exit Nano

-press **Ctrl+O**, then **Enter** to save.

-press **Ctrl+X** to exit.

- Make the script executable

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ chmod +x delete_old_logs.sh
```

- Run the script

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo ./delete_old_logs.sh
Deleted the following .log files older than 7 days:
/var/log/custom_logs/test1.log
```

-You will see a list of deleted **.log** files.

- Verify

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ls -l /var/log/custom_logs/
total 0
-rw-r--r-- 1 root root 0 @ 7 8 16:55 test1.logs
-rw-r--r-- 1 root root 0 @ 7 8 16:56 test2.log
pabodamahagedara@pabodamahagedara-VirtualBox:~$
```

-It should no longer contains files older than 7 days.

ii. Archive the remaining log files into a compressed .tar.gz file named logs_backup.tar.gz.

- Edit your script

-Reopen the script you created earlier

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ nano delete_old_logs.sh
```

- Add the archiving code below the deletion part

-update your script to look like this

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~
GNU nano 6.2                                delete_old_logs.sh *
#!/bin/bash
#task1 (i): Delete. log files older than 7 days from /var/log/custom_logs/

#define the directory
LOG_DIR="/var/log/custom_logs"
ARCHIVE_NAME="/var/log/custom_logs/logs_backup.tar.gz"

#find and delete .log files older than 7 days
DELETED_FILES=$(find "$LOG_DIR" -name "*.log" -type f -mtime +7 -print -delete)

#archive remaining .log files
sudo tar -czf "$ARCHIVE_NAME" -C "$LOG_DIR" $(sudo find "$LOG_DIR" -name "*.log" -type f)

#print summary
echo "Deleted the following .log files older than 7 days:"
echo "$DELETED_FILES"
echo "Remaining .log files have been archived to $ARCHIVE_NAME"

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-|
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-|
```


- Save and Exit Nano
- Run the script with **sudo**

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo ./delete_old_logs.sh
[sudo] password for pabodamahagedara:
tar: Removing leading '/' from member names
Deleted the following .log files older than 7 days:

Remaining .log files have been archived to /var/log/custom_logs/logs_backup.tar.gz
```

- Verify the archive file

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ls -lh /var/log/custom_logs/logs_backup.tar.gz
-rw-r--r-- 1 root root 129  8 17:48 /var/log/custom_logs/logs_backup.tar.gz
```

-You should see a compressed **.tar.gz** archive of the remaining **.log** files.

- (Optional)View contents of archive

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ tar -tzf /var/log/custom_logs/logs_backup.tar.gz
var/log/custom_logs/test2.log
pabodamahagedara@pabodamahagedara-VirtualBox:~$
```

-This list the files inside the archive.

iii. Print a summary of the deleted and archived files on the terminal.

- Open your script again

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ nano delete_old_logs.sh
```

- Replace everything with this full script

```
GNU nano 6.2 delete_old_logs.sh
#!/bin/bash

#define the directory
LOG_DIR="/var/log/custom_logs"
ARCHIVE_NAME="/var/log/custom_logs/logs_backup.tar.gz"

echo "=====
echo " Starting Log Cleanup & Backup"
echo "=====

#find and delete .log files older than 7 days
DELETE_FILES=$(sudo find "$LOG_DIR" -name "*.log" -type f -mtime +7 -print)
sudo find "$LOG_DIR" -name "*.log" -type f -mtime +7 -delete

#archive remaining .log files
REMAINING_FILES=$(sudo find "$LOG_DIR" -name "*.log" -type f)
if [ -n "$REMAINING_FILES" ]; then
    sudo tar -czf "$ARCHIVE_NAME" -C "$LOG_DIR" $(basename -a $REMAINING_FILES 2>/dev/null)
fi

#print summary
echo ""
echo "===== Summary ====="

if [ -z "$DELETED_FILES" ]; then
    echo "No .log files were deleted."
else
    echo "Deleted .log files older tan 7 days:"
    echo "$DELETED_FILES"
fi

if [ -z "$REMAINING_FILES" ]; then
    echo "No remaining .log files to archive."
else
    echo ""
    echo "Archived .log files into $ARCHIVE_NAME:"
```

```

    echo "$REMAINING_FILES"
fi

echo "======"
echo "Script Finished"
echo "======"

```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy

- Save and Exit
- Make executable

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo chmod +x delete_old_logs.sh
```

- Run with sudo

-You will see summary like below;

```

pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo ./delete_old_logs.sh
=====
Starting Log Cleanup & Backup
=====

==== Summary ====
No .log files were deleted.

Archived .log files into /var/log/custom_logs/logs_backup.tar.gz:
/var/log/custom_logs/test2.log
=====
Script Finished
=====

```

iv. Schedule this script to run automatically every Sunday at 12:00 midnight using a cron job.

- Make sure the script has execute permission

-If not already done

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ chmod +x ~/delete_old_logs.sh
```

-This ensures the script is executable.

- Move the script to a safe location

-System scripts are usually placed in **/usr/local/bin/** or **/opt/**

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo mv ~/delete_old_logs.sh /usr/local/bin/delete_old_logs.sh
```

- Edit the **crontab**

-Open **crontab** for the root user (since the script uses **sudo** operations)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo crontab -e
no crontab for root - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
```

- Add this line to the end of file

```

pabodamahagedara@pabodamahagedara-VirtualBox: ~
GNU nano 6.2 /tmp/crontab.lKSM0o/crontab
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 0 * * 0 /usr/local/bin/delete_old_logs.sh >> /var/log/log_cleanup_cron.log 2>&1

```

[Wrote 25 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-I
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-!

-Explanation;

- **0 0 * * 0** :Run at 00:00 (midnight) every Sunday
- **/usr/local/bin/delete_old_logs.sh** :Full path to your script
- **>> /var/log/log_cleanup_cron.log 2>&1** :Save output/errors to a log file.

- Save and Exit.
- (Optional) Check Cron Jobs

-To list root's cron jobs;

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 0 * * 0 /usr/local/bin/delete_old_logs.sh >> /var/log/log_cleanup_cron.log 2>&1
```

-This script will now automatically run every Sunday at midnight!

- v. Your script should be structured and commented to explain its functionality. Assume appropriate permissions are available.
- Final script: fully structured & commented.

```

GNU nano 6.2                                delete_old_logs.sh
#!/bin/bash
#-----
#script name : delete_old_logs.sh
#description : Deletes .log files older than 7 days from /var/log/custom_logs/ and archives the remaining logs into a compressed file
#author      : Paboda Mahagedara
#schedule    : set to run every sunday at 12:00 AM via crontab
#-----

#Define the directory containing the log files
LOG_DIR="/var/log/custom_logs"

#Define the name of the archive file (stored in same directory)
ARCHIVE_NAME="$LOG_DIR/logs_backup.tar.gz"

# -----

#Step 1: delete old .log files
#-----
#find and list .log files older than 7 days
DELETED_FILES=$(sudo find "$LOG_DIR" -name "*.log" -type f -mtime +7 -print)

#delete those old .log files
sudo find "$LOG_DIR" -name "*.log" -type f -mtime +7 -delete

#-----

#step 2: archived remaining .log files
#-----

#find remaining .log files
REMAINING_FILES=$(sudo find "$LOG_DIR" -name "*.log" -type f)

#create a .tar.gz archive of the remaining .log files
#-C changes to the directory before adding files to archive
#basename trims full path for cleaner archive
sudo tar -czf "$ARCHIVE_NAME" -C "$LOG_DIR" $(basename -a $REMAINING_FILES 2>/dev/null)

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute   ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify   ^_ Go To Line M-E Redo      M-6 Copy
  
```

```
#-----
#step 3: print summary report
#-----
echo ""
echo "=====
echo "log cleanup summary"
echo "=====

#if deleted files list is empty
if [ -z "$DELETED_FILES" ]; then
    echo "No .log files were deleted (none older tahn 7 days)."
else
    echo "Deleted .log files:"
    echo "$DELETED_FILES"
fi

#if no remaining files to archive
if [ -z "$REMAINING_FILES" ]; then
    echo "No .log files were left to archive."
else
    echo ""
    echo "Archived .log files into $ARCHIVE_NAME:"
    echo "$REMAINING_FILES"
fi

echo "=====
echo "script completed"
echo "===== "
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-6 Copy

- Final check before you finished

-Make sure the script save as : **/usr/local/bin/delete_old_logs.sh**

-Has executed permissions:

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo chmod +x /usr/local/bin/delete_old_logs.sh
[sudo] password for pabodamahagedara:
Sorry, try again.
[sudo] password for pabodamahagedara:
```


- Scheduled via cron (step iv)
- You can run it manually anytime:

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo /usr/local/bin/delete_old_logs.sh
=====
Starting Log Cleanup & Backup
=====

==== Summary ====
No .log files were deleted.

Archived .log files into /var/log/custom_logs/logs_backup.tar.gz:
/var/log/custom_logs/test2.log
=====
Script Finished
=====
pabodamahagedara@pabodamahagedara-VirtualBox:~$
```

3.2 SSH (Secure Shell)

-Configure and use SSH

- Open terminal
- Install the OpenSSH server (if not installed)

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~  
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt update  
[sudo] password for pabodamahagedara:  
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]  
Hit:2 http://lk.archive.ubuntu.com/ubuntu jammy InRelease  
Get:3 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]  
Get:4 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]  
Reading package lists... Done  
  
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install openssh-server  
-y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
openssh-server is already the newest version (1:8.9p1-3ubuntu0.13).  
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

-This installs the **sshd** service required for incoming SSH connections

- Check if SSH server is running

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: ena  
   Active: active (running) since Thu 2025-05-08 22:35:18 +0530; 11min ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
   Main PID: 6199 (sshd)  
     Tasks: 1 (limit: 2269)  
    Memory: 4.1M  
       CPU: 94ms  
    CGroup: /system.slice/ssh.service  
            └─6199 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
08 22:35:18 pabodamahagedara-VirtualBox systemd[1]: Stopped OpenBSD Secure >  
08 22:35:18 pabodamahagedara-VirtualBox sshd[6199]: Server listening on :: >  
08 22:35:18 pabodamahagedara-VirtualBox systemd[1]: Starting OpenBSD Secure >  
08 22:35:18 pabodamahagedara-VirtualBox systemd[1]: Started OpenBSD Secure >  
08 22:41:49 pabodamahagedara-VirtualBox sshd[66731]: Invalid user your-userna>
```

-You should see **active (running)** in green

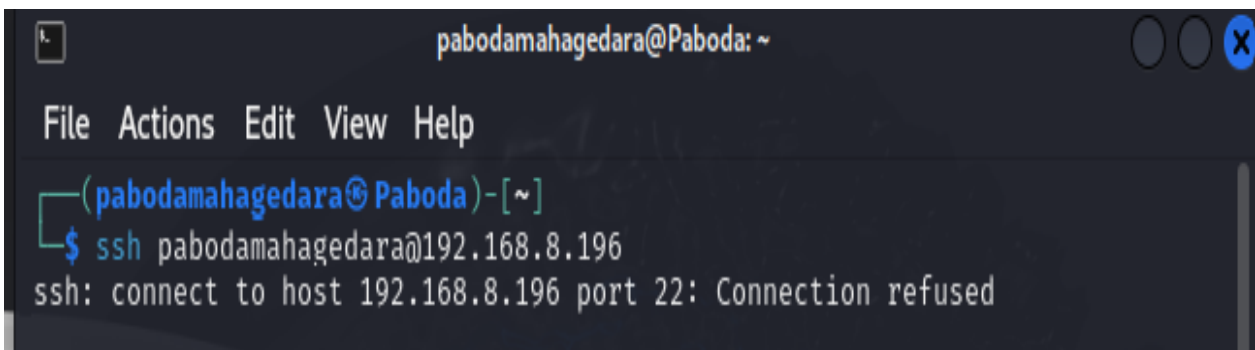
- Find the IP address of the VM

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:83:d6:ad brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fd00::3661:115a:5275:3bc0/64 scope global temporary dynamic
        valid_lft 86174sec preferred_lft 14174sec
    inet6 a00:27ff:fe83:d6ad/64 scope global dynamic mngtmpaddr
        valid_lft 86174sec preferred_lft 14174sec
    inet6 fe80::a00:27ff:fe83:d6ad/64 scope link
        valid_lft forever preferred_lft forever
```

-In there network interface (enp0s3) IP like 10.0.2.15

- Connect another computer using SSH client

-use another terminal



```
pabodamahagedara@Paboda: ~
File Actions Edit View Help
(pabodamahagedara@Paboda)-[~]
$ ssh pabodamahagedara@192.168.8.196
ssh: connect to host 192.168.8.196 port 22: Connection refused
```

-To ensure the connection between two systems, let access some files in remotely accessing system using **ls** command. This list down the directories in the remotely accessed system and display them on the system.

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ls
delete_old_logs.sh  Documents  html      Pictures  snap      Templates
Desktop             Downloads  Music     Public    snp        Videos
```

3.3 iptables and ACLs

- Open terminal
- Check your IP address

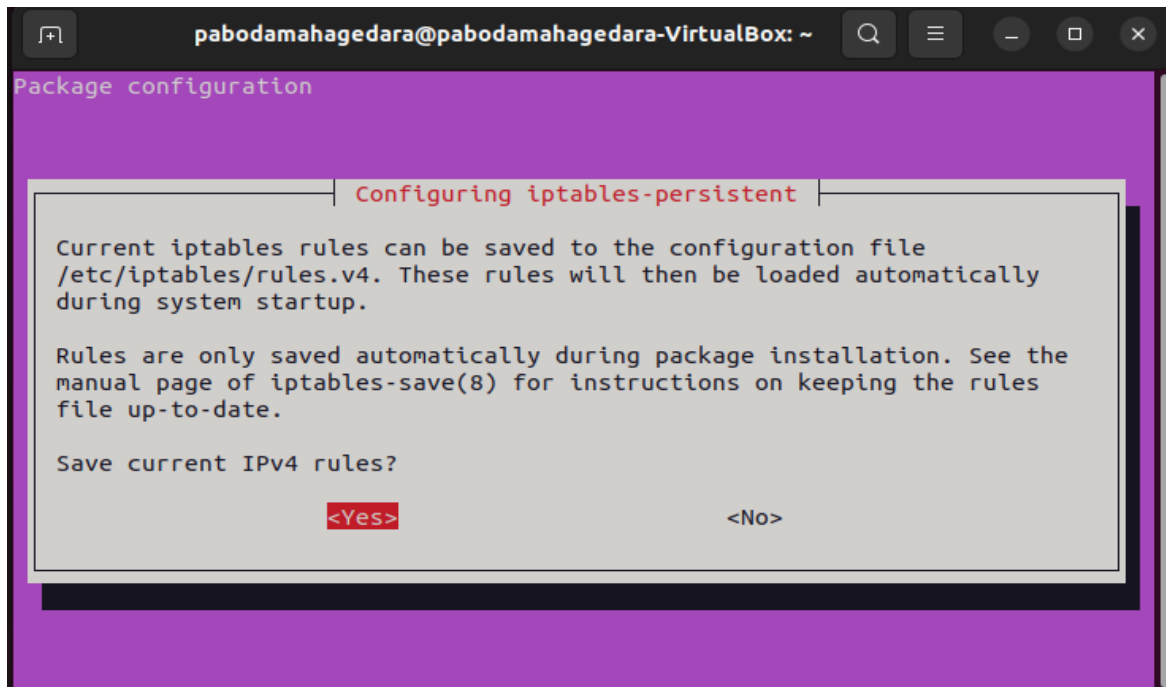
```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:83:d6:ad brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fd00::3661:115a:5275:3bc0/64 scope global temporary dynamic
        valid_lft 85986sec preferred_lft 13986sec
    inet6 fd00::a00:27ff:fe83:d6ad/64 scope global dynamic mngtmpaddr
        valid_lft 85986sec preferred_lft 13986sec
    inet6 fe80::a00:27ff:fe83:d6ad/64 scope link
        valid_lft forever preferred_lft forever
```

- Update your system and install iptables (if not already installed)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt update
[sudo] password for pabodamahagedara:
Hit:1 http://lk.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Reading package lists... Done
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install iptables iptables-persistent -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
iptables is already the newest version (1.8.7-1ubuntu5.2).
iptables set to manually installed.
The following NEW packages will be installed:
  iptables-persistent netfilter-persistent
0 upgraded, 2 newly installed, 0 to remove and 2 not upgraded.
Need to get 13.9 kB of archives.
After this operation, 93.2 kB of additional disk space will be used.
Get:1 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 netfilter-persistent all 1.0.16 [7,440 B]
Get:2 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 iptables-persistent all 1.0.16 [6,488 B]
Fetched 13.9 kB in 2s (7,945 B/s)
Preconfiguring packages ...
Selecting previously unselected package netfilter-persistent.
(Reading database ... 205168 files and directories currently installed.)
Preparing to unpack .../netfilter-persistent_1.0.16_all.deb ...
Unpacking netfilter-persistent (1.0.16) ...
Selecting previously unselected package iptables-persistent.
Preparing to unpack .../iptables-persistent_1.0.16_all.deb ...
Unpacking iptables-persistent (1.0.16) ...
Setting up netfilter-persistent (1.0.16) ...
Created symlink /etc/systemd/system/multi-user.target.wants/netfilter-persistent.service → /lib/systemd/system/netfilter-persistent.service.
Setting up iptables-persistent (1.0.16) ...
update-alternatives: using /lib/systemd/system/netfilter-persistent.service to provide /lib/systemd/system/iptables.service (iptables.service) in auto mode
Processing triggers for man-db (2.10.2-1) ...
```

-When prompted during install, choose Yes to save rules.



i. **Block Social Media Websites (Facebook, Instagram, Twitter)**

- Block URLs using iptables string match.

-This method blocks access based on the domain name appearing in packets (works on HTTP, not always HTTPS. But acceptable for basic firewall use).

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 80 -m string --string "facebook.com" --algo bm -j REJECT
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 443 -m string --string "facebook.com" --algo bm -j REJECT
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 80 -m string --string "instagram.com" --algo bm -j REJECT
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 443 -m string --string "instagram.com" --algo bm -j REJECT
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 80 -m string --string "twitter.com" --algo bm -j REJECT
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 443 -m string --string "twitter.com" --algo bm -j REJECT
```


ii. Allow Only Secure Web Browsing (HTTPS)

- Block HTTP (Port 80) and allow HTTPS (Port 443)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 80 -j REJECT
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -A OUTPUT -p tcp --dport 443 -j ACCEPT
```

- Save the iptables rules

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
```

- View and confirm rules

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo iptables -L -n --line-numbers
Chain INPUT (policy ACCEPT)
num target      prot opt source                destination

Chain FORWARD (policy ACCEPT)
num target      prot opt source                destination

Chain OUTPUT (policy ACCEPT)
num target      prot opt source                destination
1  REJECT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:80 STRING match "facebook.com" ALGO name bm TO 65535 reject-with icmp-port-unreachable
2  REJECT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:443 STRING match "facebook.com" ALGO name bm TO 65535 reject-with icmp-port-unreachable
3  REJECT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:80 STRING match "instagram.com" ALGO name bm TO 65535 reject-with icmp-port-unreachable
4  REJECT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:443 STRING match "instagram.com" ALGO name bm TO 65535 reject-with icmp-port-unreachable
5  REJECT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:80 STRING match "twitter.com" ALGO name bm TO 65535 reject-with icmp-port-unreachable
6  REJECT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:443 STRING match "twitter.com" ALGO name bm TO 65535 reject-with icmp-port-unreachable
7  REJECT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:80 reject-with icmp-port-unreachable
8  ACCEPT        tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:443
```

- Test firewall rules

1. Test HTTP Block

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ curl http://example.com
```

-In there it blinks a few seconds.

-It should fail or get reject.

2. Test HTTPS works

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~  
pabodamahagedara@pabodamahagedara-VirtualBox:~$ curl https://example.com  
<!doctype html>  
<html>  
<head>  
  <title>Example Domain</title>  
  
  <meta charset="utf-8" />  
  <meta http-equiv="Content-type" content="text/html; charset=utf-8" />  
  <meta name="viewport" content="width=device-width, initial-scale=1" />  
  <style type="text/css">  
    body {  
      background-color: #f0f0f2;  
      margin: 0;  
      padding: 0;  
      font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "Open Sans", "Helvetica Neue", Helvetica, Arial, sans  
-serif;  
    }  
    div {  
      width: 600px;  
      margin: 5em auto;  
      padding: 2em;  
      background-color: #fdfdff;  
      border-radius: 0.5em;  
      box-shadow: 2px 3px 7px 2px rgba(0,0,0,0.02);  
    }  
    a:link, a:visited {  
      color: #38488f;  
      text-decoration: none;  
    }  
    @media (max-width: 700px) {  
      div {  
        margin: 0 auto;  
        width: auto;  
      }  
    }  
  </style>  
</head>  
<body>
```


-This should work.

3. Test Facebook block

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ curl https://facebook.com
curl: (35) OpenSSL SSL_connect: Connection reset by peer in connection to facebook.com:443
```

-Should be blocked or hang.

3.4 Web Server (Apache/Nginx)

-Install & configure Apache web server

- Install Apache

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt update
[sudo] password for pabodamahagedara:
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://lk.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://lk.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://lk.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Reading package lists... Done
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  grub-efi-amd64-bin grub-efi-amd64-signed
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
0 upgraded, 8 newly installed, 0 to remove and 2 not upgraded.
Need to get 1,922 kB of archives.
After this operation, 7,728 kB of additional disk space will be used.
Get:1 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9,170 B]
Get:5 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.14 [1,349 kB]
Get:6 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.14 [165 kB]
Get:7 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.14 [89.0 kB]
Get:8 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.14 [97.9 kB]
Fetched 1,922 kB in 6s (333 kB/s)
Selecting previously unselected package libapr1:amd64.
(Reading database ... 205197 files and directories currently installed.)
Preparing to unpack .../0-libapr1_1.7.0-8ubuntu0.22.04.2_amd64.deb ...
Unpacking libapr1:amd64 (1.7.0-8ubuntu0.22.04.2) ...
Selecting previously unselected package libaprutil1:amd64.
Preparing to unpack .../1-libaprutil1_1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../3-libaprutil1-ldap_1.6.1-5ubuntu4.22.04.2_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../4-apache2-bin_2.4.52-1ubuntu4.14_amd64.deb ...
Unpacking apache2-bin (2.4.52-1ubuntu4.14) ...
Selecting previously unselected package apache2-data.
Preparing to unpack .../5-apache2-data_2.4.52-1ubuntu4.14_all.deb ...
Unpacking apache2-data (2.4.52-1ubuntu4.14) ...
Selecting previously unselected package apache2-utils.
Preparing to unpack .../6-apache2-utils_2.4.52-1ubuntu4.14_amd64.deb ...
Unpacking apache2-utils (2.4.52-1ubuntu4.14) ...
Selecting previously unselected package apache2.
Preparing to unpack .../7-apache2_2.4.52-1ubuntu4.14_amd64.deb ...
Unpacking apache2 (2.4.52-1ubuntu4.14) ...
```

- Start and Enable Apache

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
```

- Create simple web page

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /var/www/html/index.html
```



```
GNU nano 6.2 /var/www/html/index.html *
<!DOCTYPE html>
<html>
<head>
  <title>My First Web server</title>
</head>
<body>
  <h1> Success! Your Apache server is working! </h1>
</body>
</html>

File Name to Write: /var/www/html/index.html
^G Help          M-D DOS Format    M-A Append
^C Cancel        M-M Mac Format    M-P Prepend
```

- Allow HTTP through the firewall

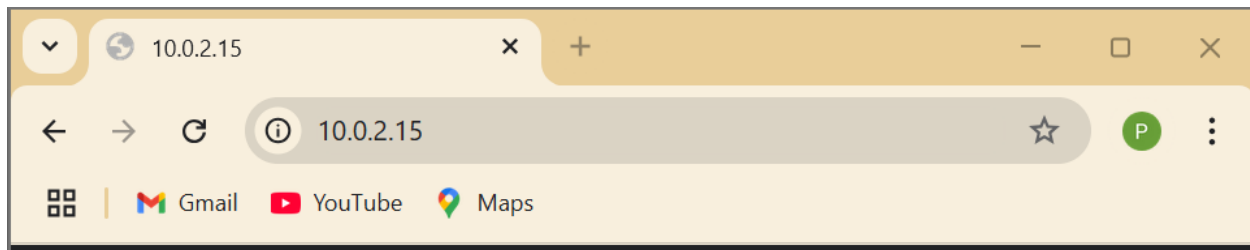
```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo ufw allow 'Apache'
Rules updated
Rules updated (v6)
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
```

- Find your VM's IP address

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:83:d6:ad brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fd00::3661:115a:5275:3bc0/64 scope global temporary dynamic
        valid_lft 85990sec preferred_lft 13990sec
    inet6 fd00::a00:27ff:fe83:d6ad/64 scope global dynamic mngtmpaddr
        valid_lft 85990sec preferred_lft 13990sec
    inet6 fe80::a00:27ff:fe83:d6ad/64 scope link
        valid_lft forever preferred_lft forever
pabodamahagedara@pabodamahagedara-VirtualBox:~$
```

- Access the web page

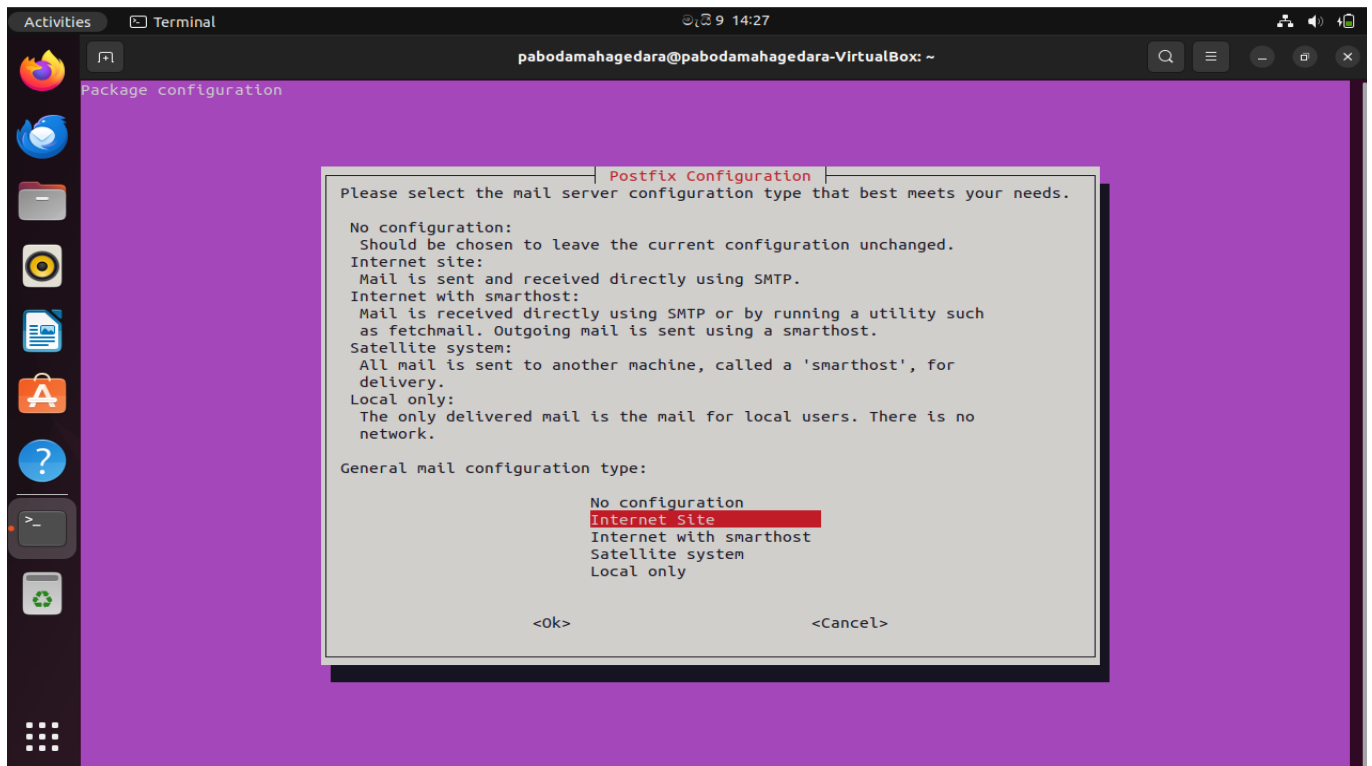


3.5 Email Server (postfix/Exim)

- Install Postfix

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~  
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install postfix -y  
[sudo] password for pabodamahagedara:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Suggested packages:  
  procmail postfix-mysql postfix-pgsql postfix-ldap postfix-pcre postfix-lmdb postfix-sqlite sasl2-bin | dovecot-common resolvconf  
  postfix-cdb postfix-mta-sts-resolver postfix-doc  
The following NEW packages will be installed:  
  postfix  
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.  
Need to get 1,248 kB of archives.  
After this operation, 4,182 kB of additional disk space will be used.  
Get:1 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 postfix amd64 3.6.4-1ubuntu1.3 [1,248 kB]  
Fetched 1,248 kB in 3s (398 kB/s)  
Preconfiguring packages ...  
Selecting previously unselected package postfix.  
(Reading database ... 205947 files and directories currently installed.)  
Preparing to unpack .../postfix_3.6.4-1ubuntu1.3_amd64.deb ...  
Unpacking postfix (3.6.4-1ubuntu1.3) ...  
Setting up postfix (3.6.4-1ubuntu1.3) ...  
Adding group `postfix' (GID 140) ...  
Done.  
Adding system user `postfix' (UID 133) ...  
Adding new user `postfix' (UID 133) with group `postfix' ...  
Not creating home directory `/var/spool/postfix'.  
Creating /etc/postfix/dynamicmaps.cf  
Adding group `postdrop' (GID 141) ...  
Done.  
setting myhostname: pabodamahagedara-VirtualBox  
setting alias maps  
setting alias database  
mailname is not a fully qualified domain name. Not changing /etc/mailname.  
setting destinations: $myhostname, pabodamahagedara-VirtualBox, localhost.localdomain, , localhost  
setting relayhost:  
setting mynetworks: 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128  
setting mailbox_size_limit: 0  
setting recipient_delimiter: +  
setting inet_interfaces: all  
setting inet_protocols: all
```

-During installation, you will see a purple screen for configuration:



-System mail name: set as your VM hostname or local host (like pabodamahagedara-VirtualBox)

- Configure Postfix (Optional but Good)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo nano /etc/postfix/main.cf
```

-Check and set;

```

GNU nano 6.2 /etc/postfix/main.cf *
#delay_warning_time = 4h

readme_directory = no

# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 3.6 on
# fresh installs.
compatibility_level = 3.6

# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache

smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = pabodamahagedara-VirtualBox
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
mydestination = $myhostname, pabodamahagedara-VirtualBox, localhost.localdomain, , localhost
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all

myhostname = localhost
mydestination = $myhostname, localhost.localdomain, localhost
inet_interfaces = all

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Un
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Re

```

- Save and Exit

- Restart Postfix

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo systemctl restart postfix
```

- Test sending a local Email

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~  
pabodamahagedara@pabodamahagedara-VirtualBox:~$ sudo apt install mailutils -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  gssasl-common guile-3.0-libs libgssasl7 libmailutils8 libmysqlclient21 libntlm0 libpq5 mailutils-common mysql-common  
Suggested packages:  
  mailutils-mh mailutils-doc  
The following NEW packages will be installed:  
  gssasl-common guile-3.0-libs libgssasl7 libmailutils8 libmysqlclient21 libntlm0 libpq5 mailutils mailutils-common mysql-common  
0 upgraded, 10 newly installed, 0 to remove and 2 not upgraded.  
Need to get 10.2 MB of archives.  
After this operation, 66.2 MB of additional disk space will be used.  
Get:1 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 gssasl-common all 1.10.0-5 [59.8 kB]  
Get:2 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 guile-3.0-libs amd64 3.0.7-1 [7,538 kB]  
Get:3 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 libntlm0 amd64 1.6-4 [73.7 kB]  
Get:4 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 libgssasl7 amd64 1.10.0-5 [71.1 kB]  
Get:5 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 mailutils-common all 1:3.14-1 [382 kB]  
Get:6 http://lk.archive.ubuntu.com/ubuntu jammy/main amd64 mysql-common all 5.8+1.0.8 [7,212 B]  
Get:7 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libmysqlclient21 amd64 8.0.42-0ubuntu0.22.04.1 [1,308 kB]  
Get:8 http://lk.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpq5 amd64 14.17-0ubuntu0.22.04.1 [152 kB]  
Get:9 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 libmailutils8 amd64 1:3.14-1 [519 kB]  
Get:10 http://lk.archive.ubuntu.com/ubuntu jammy/universe amd64 mailutils amd64 1:3.14-1 [130 kB]  
Fetched 10.2 MB in 9s (1,182 kB/s)  
Selecting previously unselected package gssasl-common.  
(Reading database ... 206144 files and directories currently installed.)  
Preparing to unpack .../0-gssasl-common_1.10.0-5_all.deb ...  
Unpacking gssasl-common (1.10.0-5) ...  
Selecting previously unselected package guile-3.0-libs:amd64.  
Preparing to unpack .../1-guile-3.0-libs_3.0.7-1_amd64.deb ...  
Unpacking guile-3.0-libs:amd64 (3.0.7-1) ...  
Selecting previously unselected package libntlm0:amd64.  
Preparing to unpack .../2-libntlm0_1.6-4_amd64.deb ...  
Unpacking libntlm0:amd64 (1.6-4) ...  
Selecting previously unselected package libgssasl7:amd64.  
Preparing to unpack .../3-libgssasl7_1.10.0-5_amd64.deb ...  
Unpacking libgssasl7:amd64 (1.10.0-5) ...  
Selecting previously unselected package mailutils-common.  
Preparing to unpack .../4-mailutils-common_1%3a3.14-1_all.deb ...  
Unpacking mailutils-common (1:3.14-1) ...
```

- Send an email

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ whoami
pabodamahagedara
pabodamahagedara@pabodamahagedara-VirtualBox:~$ echo "Test email from Postfix" | mail -s "Hello Postfix" pabodamahagedara@localhost
```

- View mail

```
pabodamahagedara@pabodamahagedara-VirtualBox:~$ mail
"/var/mail/pabodamahagedara": 1 message 1 new
>N 1 PabodaMahagedara  88  14/528  Hello Postfix
? █
```

-In there I used first method of following below methods;

Use:

- **Enter** to read
- **q** to quit
- **d** to delete

4. Linux GDB

4.a Execution Process

-Select appropriate Executable

- Open terminal
- Run below command for the verify architecture

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads$ uname -m  
x86_64
```

Based on the output, we can select the correct executable from zip file (like x86_64);

- x86_64 – 64-bit executable
- i686 - 32-bit executable
- arm, aarch64 – ARM executable
 - Download the zip file.
 - Navigate to the directory containing the executables

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads$ cd ~/Downloads
```

- Extract the zip file if not done

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads$ unzip 'Executables (1)' -d gdb_task  
Archive: Executables (1).zip  
creating: gdb_task/Executables/  
inflating: gdb_task/Executables/ARM  
inflating: gdb_task/___MACOSX/Executables/._ARM  
inflating: gdb_task/Executables/x86_64  
inflating: gdb_task/___MACOSX/Executables/._x86_64
```

- Go to the executable folder

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads$ cd gdb_task
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task$ cd Executables
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ls
ARM  x86_64
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ls
ARM  x86_64
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ls -l
total 88
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 70824 17 06:26 ARM
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 16352 17 06:26 x86_64
```

-This confirms x86_64 is an executable file, not a folder.

- Choose and run the executable

-Rename the file for clarity (Optional but helpful)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ mv x86_64 program
```

-Give the execution permission

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ chmod +x program
```

-Run it with sudo;

When you running, it will ask you to enter your IT number, enter it like below;

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ sudo ./program
Enter the student IT number: IT23623736
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$
```

- Run the genrate executable

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ chmod +x IT23623736
chmod: changing permissions of 'IT23623736': Operation not permitted
```

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ./IT23623736
```

-It will now generate data.txt

- Inspect **data.txt**

-verify the file was created,

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ls -l
total 124
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 519  9 16:25 after.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 70824 17 06:26 ARM
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 433  9 16:23 before.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 1  9 16:24 data.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 0  9 16:37 decoded.txt
-rwxr-xr-x 1 root root 20392  9 15:46 IT23623736
-rwxrwxr-x 1 pabodamahagedara pabodamahagedara 16352 17 06:26 program
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 397  9 16:26 stat.txt
```

- Initial Analysis

-List and view the file;

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ cat data.txt
```

-If it looks unreadable, try;

- Use string to extract printable text;

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ strings data.txt
```

- Use hexdump to view raw byte format;

```
things 0000.txt
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ h
exdump -C data.txt
00000000 5b                                     |[]
00000001
```

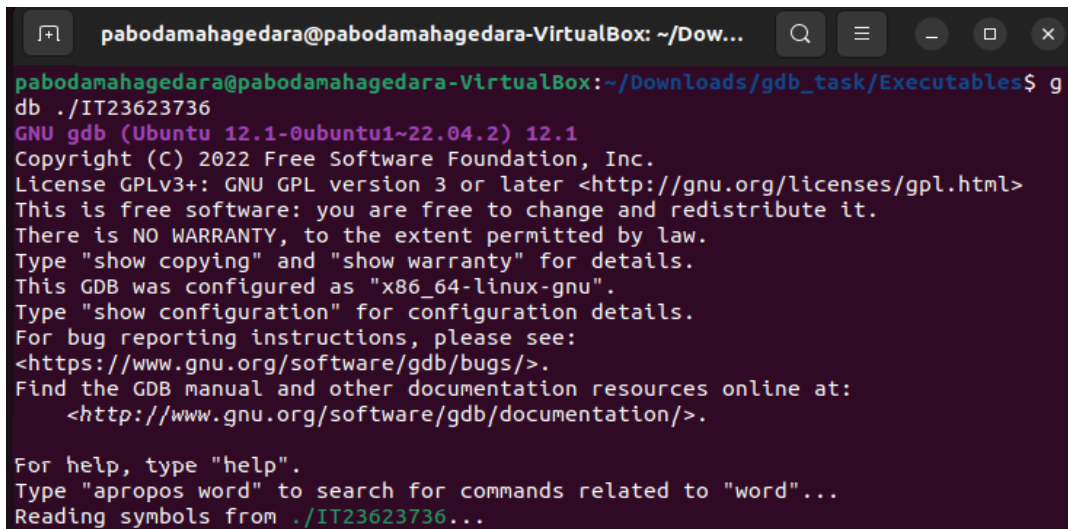
- Use file to check file type;

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ f
ile data.txt
data.txt: very short file (no magic)
```

4.b Debugging Process

i. Step-by-step debugging using GDB

- Start GDB with compiled binary



```
pabodamahagedara@pabodamahagedara-VirtualBox: ~/Dow...
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ g
db ./IT23623736
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
  <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./IT23623736...
```

- ✓ Checking for source code

```
(gdb) list
1      IT23623736.c: No such file or directory.
(gdb) █
```

- In there using list to check if the source code is available
- But in there no any file or directory.

✓ Checking for functions

```
(gdb) info functions
All defined functions:

File IT23623736.c:
13:      int main();
5:      void xor_encrypt_decrypt(char *, const char *);

Non-debugging symbols:
0x0000000000000100  _init
0x000000000000010d  __cxa_finalize@plt
0x000000000000010e  puts@plt
0x000000000000010f  fclose@plt
0x0000000000000110  strlen@plt
0x0000000000000111  __stack_chk_fail@plt
0x0000000000000112  pclose@plt
0x0000000000000113  fputs@plt
0x0000000000000114  strcspn@plt
0x0000000000000115  fgets@plt
0x0000000000000116  popen@plt
0x0000000000000117  fopen@plt
0x0000000000000118  _start
0x000000000000011b  deregister_tm_clones
0x000000000000011e  register_tm_clones
0x0000000000000120  do_global_ctors_aux
```

- In there I can found two functions, which **int main();** and **void xor_encrypt_decrypt(char *, const char *).**
- So, now we can identify the content of data.txt being encrypted might be correct and since this function uses XOR to encrypt and decrypt processes must be the same.

✓ Disassembling main function

```
(gdb) disassemble main
Dump of assembler code for function main:
0x00000000000012ee <+0>:    endbr64
0x00000000000012f2 <+4>:    push    %rbp
0x00000000000012f3 <+5>:    mov     %rsp,%rbp
0x00000000000012f6 <+8>:    sub     $0x60,%rsp
0x00000000000012fa <+12>:   mov     %fs:0x28,%rax
0x0000000000001303 <+21>:   mov     %rax,-0x8(%rbp)
0x0000000000001307 <+25>:   xor     %eax,%eax
0x0000000000001309 <+27>:   lea     0xcf8(%rip),%rax      # 0x2008
0x0000000000001310 <+34>:   mov     %rax,%rsi
0x0000000000001313 <+37>:   lea     0xcf6(%rip),%rax      # 0x2010
0x000000000000131a <+44>:   mov     %rax,%rdi
0x000000000000131d <+47>:   call    0x1160 <popen@plt>
0x0000000000001322 <+52>:   mov     %rax,-0x58(%rbp)
0x0000000000001326 <+56>:   cmpq    $0x0,-0x58(%rbp)
0x000000000000132b <+61>:   jne     0x1346 <main+88>
0x000000000000132d <+63>:   lea     0xd04(%rip),%rax      # 0x2038
0x0000000000001334 <+70>:   mov     %rax,%rdi
0x0000000000001337 <+73>:   call    0x10e0 <puts@plt>
0x000000000000133c <+78>:   mov     $0x1,%eax
0x0000000000001341 <+83>:   jmp     0x13fe <main+272>
0x0000000000001346 <+88>:   mov     -0x58(%rbp),%rdx
0x000000000000134a <+92>:   lea     -0x40(%rbp),%rax
--Type <RET> for more, q to quit, c to continue without paging--<RET>
```

```
--Type <RET> for more, q to quit, c to continue without paging--<RET>
0x000000000000134e <+96>:   mov     $0x32,%esi
0x0000000000001353 <+101>:  mov     %rax,%rdi
0x0000000000001356 <+104>:  call    0x1150 <fgets@plt>
0x000000000000135b <+109>:  mov     -0x58(%rbp),%rax
0x000000000000135f <+113>:  mov     %rax,%rdi
0x0000000000001362 <+116>:  call    0x1120 <pclose@plt>
0x0000000000001367 <+121>:  lea     -0x40(%rbp),%rax
0x000000000000136b <+125>:  lea     0xcdc(%rip),%rdx      # 0x204e
0x0000000000001372 <+132>:  mov     %rdx,%rsi
0x0000000000001375 <+135>:  mov     %rax,%rdi
0x0000000000001378 <+138>:  call    0x1140 <strncpy@plt>
0x000000000000137d <+143>:  movb    $0x0,-0x40(%rbp,%rax,1)
0x0000000000001382 <+148>:  lea     0xcc7(%rip),%rax      # 0x2050
0x0000000000001389 <+155>:  mov     %rax,-0x50(%rbp)
0x000000000000138d <+159>:  mov     -0x50(%rbp),%rdx
0x0000000000001391 <+163>:  lea     -0x40(%rbp),%rax
0x0000000000001395 <+167>:  mov     %rdx,%rsi
0x0000000000001398 <+170>:  mov     %rax,%rdi
0x000000000000139b <+173>:  call    0x1269 <xor_encrypt_decrypt>
0x00000000000013a0 <+178>:  lea     0xcad(%rip),%rax      # 0x2054
0x00000000000013a7 <+185>:  mov     %rax,%rsi
0x00000000000013aa <+188>:  lea     0xca5(%rip),%rax      # 0x2056
0x00000000000013b1 <+195>:  mov     %rax,%rdi
--Type <RET> for more, q to quit, c to continue without paging--
```



```

0x00000000000013b4 <+198>: call 0x1170 <fopen@plt>
0x00000000000013b9 <+203>: mov  %rax,-0x48(%rbp)
0x00000000000013bd <+207>: cmpq $0x0,-0x48(%rbp)
0x00000000000013c2 <+212>: jne 0x13da <main+236>
0x00000000000013c4 <+214>: lea 0xc94(%rip),%rax # 0x205f
0x00000000000013cb <+221>: mov  %rax,%rdi
0x00000000000013ce <+224>: call 0x10e0 <puts@plt>
0x00000000000013d3 <+229>: mov  $0x1,%eax
0x00000000000013d8 <+234>: jmp 0x13fe <main+272>
0x00000000000013da <+236>: mov  -0x48(%rbp),%rdx
0x00000000000013de <+240>: lea  -0x40(%rbp),%rax
0x00000000000013e2 <+244>: mov  %rdx,%rsi
0x00000000000013e5 <+247>: mov  %rax,%rdi
0x00000000000013e8 <+250>: call 0x1130 <fputs@plt>
0x00000000000013ed <+255>: mov  -0x48(%rbp),%rax
0x00000000000013f1 <+259>: mov  %rax,%rdi
0x00000000000013f4 <+262>: call 0x10f0 <fclose@plt>
0x00000000000013f9 <+267>: mov  $0x0,%eax
0x00000000000013fe <+272>: mov  -0x8(%rbp),%rdx
0x0000000000001402 <+276>: sub  %fs:0x28,%rdx
0x000000000000140b <+285>: je   0x1412 <main+292>
0x000000000000140d <+287>: call 0x1110 <__stack_chk_fail@plt>
0x0000000000001412 <+292>: leave
0x0000000000001413 <+293>: ret
End of assembler dump.
(gdb)

```

ii. Setting breakpoints

- I analyzed to put breakpoints to the important function called places;

1. Before popen()

```
0x000000000000131d <+47>: call 0x1160 <popen@plt>
```

2. Before fgets()

```
0x0000000000001356 <+104>: call 0x1150 <fgets@plt>
```

3. Before strcspn()

```
0x0000000000001378 <+138>: call 0x1140 <strcspn@plt>
```

4. Before xor_encrypt_decrypt()

```
0x000000000000139b <+173>: call 0x1269 <xor_encrypt_decrypt>
```

5. Before fopen()

```
0x00000000000013b4 <+198>: call 0x1170 <fopen@plt>
```

6. Before fputs()

```
0x00000000000013e8 <+250>: call 0x1130 <fputs@plt>
```

- In there, I tried to add breakpoint using relative addresses and tried to run

```
(gdb) break *(main+47)
Breakpoint 12 at 0x5555555531d: file IT23623736.c, line 14.
(gdb) run
```

```
Breakpoint 12, 0x00005555555531d in main () at IT23623736.c:14
14      in IT23623736.c
```

- This output shows it is successfully attempted
- Then I added the reset of the breakpoints using relative addresses

```
(gdb) break *(main+104)
Breakpoint 13 at 0x55555555356: file IT23623736.c, line 20.
(gdb) break *(main+138)
Breakpoint 14 at 0x55555555378: file IT23623736.c, line 23.
(gdb) break *(main+173)
Breakpoint 15 at 0x5555555539b: file IT23623736.c, line 26.
(gdb) break *(main+198)
Breakpoint 16 at 0x555555553b4: file IT23623736.c, line 27.
(gdb) break *(main+250)
Breakpoint 17 at 0x555555553e8: file IT23623736.c, line 32.
(gdb) █
```

- Then I can confirm breakpoints are successfully added or not by using **info breakpoints**

```
(gdb) info breakpoints
Num    Type             Disp Enb Address                  What
12     breakpoint       keep y   0x00005555555531d in main at IT23623736.c:14
       breakpoint already hit 1 time
13     breakpoint       keep y   0x000055555555356 in main at IT23623736.c:20
14     breakpoint       keep y   0x000055555555378 in main at IT23623736.c:23
15     breakpoint       keep y   0x00005555555539b in main at IT23623736.c:26
16     breakpoint       keep y   0x0000555555553b4 in main at IT23623736.c:27
17     breakpoint       keep y   0x0000555555553e8 in main at IT23623736.c:32
(gdb) █
```

-Run code again

```
(gdb) run
The program being debugged has been started already.
Start it from the beginning? (y or n) y
Starting program: /home/pabodamahagedara/Downloads/gdb_task/Executables/IT23623736
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 12, 0x000055555555531d in main () at IT23623736.c:14
14      in IT23623736.c
(gdb) █
```

4.c File system Analysis

4.c.ii Function analysis

- Analyzing the first breakpoints (line 14)

```
Breakpoint 1, 0x00005555555531d in main () at IT23623736.c:14
14      IT23623736.c: No such file or directory.
```

-The debugger (gdb) is paused at line 14 in the main() function of your source file (IT23623736.c)

-The warning indicates that the debugger can't find the source file, but execution continues in the compiled binary

```
(gdb) step
_IO_new_popen (command=0x555555556010 "sudo cat /sys/class/dmi/id/product_uuid",
mode=0x555555556008 "r") at ./libio/iopopen.c:220
220      ./libio/iopopen.c: No such file or directory.
```

-You use the **step** command at the line calling **popen()**, GDB tries to step into the implementation of the **popen()** function.

-Can see arguments like,

- `command` is the shell command being executed (e.g., `"sudo cat /sys/class/dmi/id/product_uuid"`).
- `mode` is the mode string (`"r"` for reading)

-Since **popen()** is part of the C standard library (glibc), GDB attempts to load its source code. However, unless you have the glibc source installed, GDB cannot find it and you see a warning:

./libio/iopopen.c: No such file or directory.

-This warning is **normal** and simply means GDB cannot show you the source code for the library function

-If you want to **skip over** the internal library code and return to your own code, use the **finish** command:

```
(gdb) finish
Run till exit from #0  _IO_new_popen (
  command=0x555555556010 "sudo cat /sys/class/dmi/id/product_uuid",
  mode=0x555555556008 "r") at ./libio/iopopen.c:220
[Detaching after vfork from child process 20057]
0x000055555555322 in main () at IT23623736.c:14
14      IT23623736.c: No such file or directory.
Value returned is $1 = (FILE *) 0x55555555592a0
(gdb) █
```

-The finish command continues execution until the current function (popen()) completes and returns control to the calling code.

- The debugger resumes execution inside the internal implementation of popen() and runs until the function finishes.
 - Upon the return, GDB displays,
- The command and mode used for popen():

```
command=0x555555556010 "sudo cat /sys/class/dmi/id/product_uuid",
mode=0x555555556008 "r") at ./libio/iopopen.c:220
```

- The location in the source code where execution resumes (in this case, line 14 of IT23623736.c)
- The value returned by popen() :

```
Value returned is $1 = (FILE *) 0x55555555592a0
```

- This is a pointer to a FILE object, which represents the pipe connected to the output of the shell command.

- Analyzing the second breakpoint (line 20)

```
Breakpoint 2, 0x0000555555555356 in main () at IT23623736.c:20
20      in IT23623736.c
```

-The debugger has stopped at line 20, ready to execute the the fgets() function.

```
(gdb) step
_IO_fgets (buf=0x7fffffffde20 "", n=50, fp=0x555555592a0) at ./libio/iofgets.c:
32
32      ./libio/iofgets.c: No such file or directory.
```

-Internal implementation of fgets() with the debugging steps;

- buf is the buffer to store the read string.
- n=50 specifies the maximum number of characters to read.
- fp is the file pointer (likely from popen()).

```
(gdb) finish
Run till exit from #0 _IO_fgets (buf=0x7fffffffde20 "", n=50,
fp=0x555555592a0) at ./libio/iofgets.c:32
main () at IT23623736.c:21
21      IT23623736.c: No such file or directory.
Value returned is $2 = 0x7fffffffde20 "0e558f52-33f7-294b-94fd-7e72ae383975\n"
(gdb) break *(main+128)
```

-The finish command completes the execution of fgets().

- The output shows the program has returned from internal _IO_fgets function back to source code at line 21.
- The buffer now contains the string "0e558f52-33f7-294b-94fd-7e72ae383975\n", which is the value read from the file or command output.

- The value returned is the memory address of buffer, confirming that fgets() successfully read the string.
- The warning about missing source files (IT23623736.c: No such file or directory) is normal and does not affect to debugging session.

- **Analyzing the third breakpoint (line 23)**

```
Breakpoint 3 at 0x55555555378: file IT23623736.c, line 23.
```

The debugger has stopped at line 23 , analyze the fclose() call;

```
(gdb) step
_IO_new_fclose (fp=0x5555555592a0) at ./libio/iofclose.c:34
34      ./libio/iofclose.c: No such file or directory.
```

-The debugger steps into the internal implementation of fclose() function.

- fp is the file pointer to be closed (here, 0x5555555592a0)
- The function _IO_new_fclose is part of the C library's file handling implementation.
- The warning about missing source files is normal and indicates that GDB can not find the source code for the standard library, not your program.

```
(gdb) finish
Run till exit from #0 _IO_new_fclose (fp=0x5555555592a0)
  at ./libio/iofclose.c:34
main () at IT23623736.c:23
23      IT23623736.c: No such file or directory.
Value returned is $3 = 0
```

-The finish command of completes the execution of fclose() function.

- The program returns from _IO_new_fclose back to your source code at line 23 .

- The value returned is $\$3 = 0$, which means the file was successfully closed.

4.d Analysis of "data.txt"

d.ii. Source of the contents of data.txt

i. Observations on file system changes

- ✓ Before execution

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ls -l > before.txt
```

-Then also run

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ stat * > metadata_before.txt
```

-Output or saved file showing metadata (metadata_before.txt)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ls -l
total 128
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 519  9 16:25 after.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 70824 17 06:26 ARM
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 692  9 19:59 before.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 1  9 16:24 data.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 0  9 16:37 decoded.txt
-rwxr-xr-x 1 root root 20392  9 15:46 IT23623736
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 3168  9 19:59 metadata_before.txt
-rwxrwxr-x 1 pabodamahagedara pabodamahagedara 16352 17 06:26 program
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 397  9 16:26 stat.txt
```

- ✓ Run the executable again


```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Execu
[sudo] password for pabodamahagedara:
```

-let the program complete its run. It will likely re-generate or overwrite **data.txt** .

✓ List files again after execution

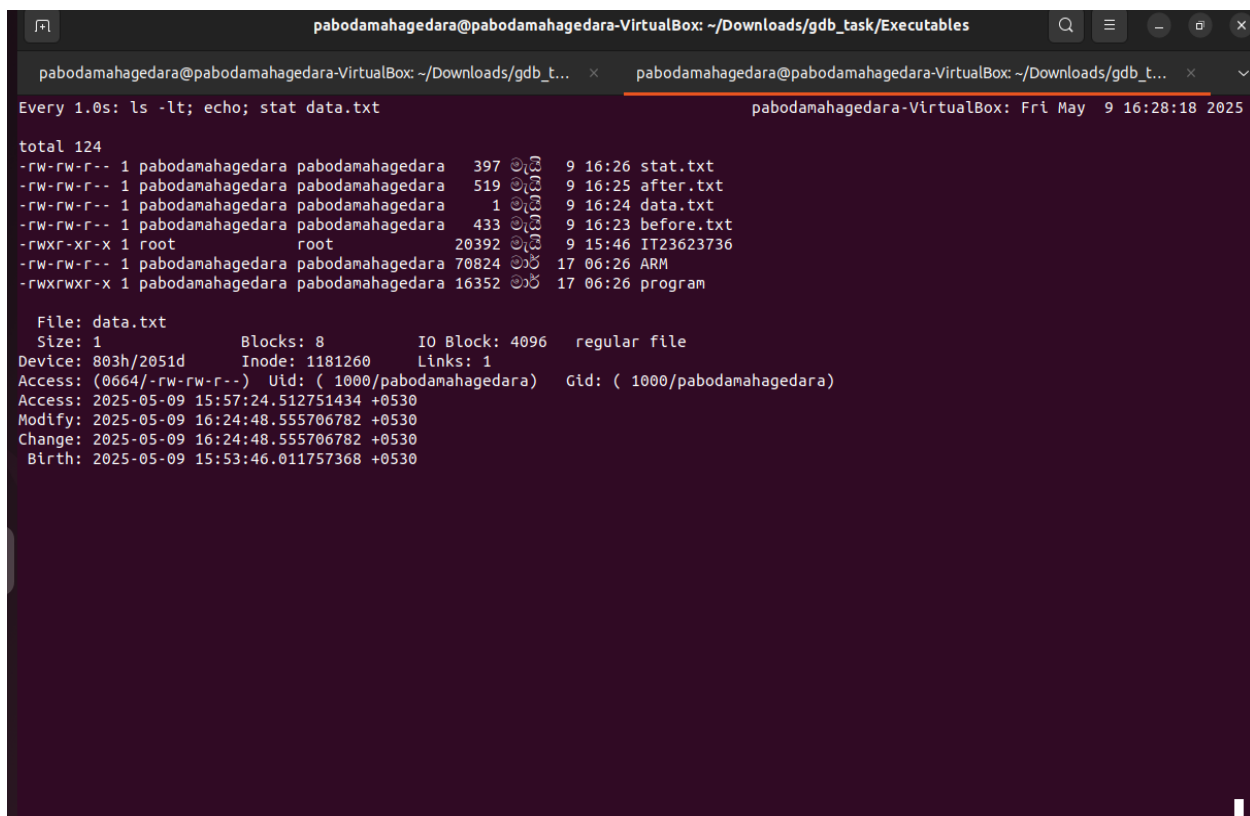
```
[sudo] password for pabodamahagedara:
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ ls -l > after.txt
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ stat * > metadata_after.txt
```

-Now compare before and after

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ diff before.txt after.txt
1,2c1,2
< total 120
< -rw-rw-r-- 1 pabodamahagedara pabodamahagedara  519  9 16:25 after.txt
---
> total 124
> -rw-rw-r-- 1 pabodamahagedara pabodamahagedara    0  9 20:11 after.txt
4,5c4,5
< -rw-rw-r-- 1 pabodamahagedara pabodamahagedara    0  9 19:59 before.txt
< -rw-rw-r-- 1 pabodamahagedara pabodamahagedara    1  9 16:24 data.txt
---
> -rw-rw-r-- 1 pabodamahagedara pabodamahagedara  692  9 19:59 before.txt
> -rw-rw-r-- 1 pabodamahagedara pabodamahagedara    1  9 20:06 data.txt
7a8
> -rw-rw-r-- 1 pabodamahagedara pabodamahagedara  3168  9 19:59 metadata_before.txt
```

- ✓ Screenshots showing file modifications

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~/Downloads/gdb_task/Executables$ watch -n 1 "ls -lt; echo; stat data.txt"
```



The screenshot shows a terminal window titled "pabodamahagedara@pabodamahagedara-VirtualBox: ~/Downloads/gdb_task/Executables". The terminal displays the output of the command `watch -n 1 "ls -lt; echo; stat data.txt"`. The output shows a list of files in the directory, with the file `data.txt` being monitored. The output includes the file's permissions, owner, size, and modification time. Below the file list, the output of the `stat` command for `data.txt` is shown, including details like file type, size, blocks, IO block, device, inode, links, access, modification, change, and birth times.

```
Every 1.0s: ls -lt; echo; stat data.txt                                pabodamahagedara-VirtualBox: Fri May  9 16:28:18 2025

total 124
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 397  9 16:26 stat.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 519  9 16:25 after.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara  1  9 16:24 data.txt
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 433  9 16:23 before.txt
-rwxr-xr-x 1 root             root             20392  9 15:46 IT23623736
-rw-rw-r-- 1 pabodamahagedara pabodamahagedara 70824 17 06:26 ARM
-rwxrwxr-x 1 pabodamahagedara pabodamahagedara 16352 17 06:26 program

  File: data.txt
  Size: 1          Blocks: 8          IO Block: 4096   regular file
Device: 803h/2051d Inode: 1181260    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/pabodamahagedara)   Gid: ( 1000/pabodamahagedara)
Access: 2025-05-09 15:57:24.512751434 +0530
Modify: 2025-05-09 16:24:48.555706782 +0530
Change: 2025-05-09 16:24:48.555706782 +0530
 Birth: 2025-05-09 15:53:46.011757368 +0530
```

- ✓ Use strings and hexdump to analyze file contents

-To extract human-readable text:

```
pabodamahagedara@pabodamahagedara-VirtualBox: ~/Downloads/gdb_task/Executables$ strings data.txt
```

-To view binary layout:

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ hexdump -C data.txt
00000000  5b                                | |
00000001
```

✓ Use GDB to trace functions and variables

-start GDB

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ gdb ./IT23623736
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
  <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./IT23623736...
```

-Set breakpoints at file writing functions or which functions, identified earlier

```
(gdb) break main
Breakpoint 1 at 0x12fa: file IT23623736.c, line 13.
(gdb) run
Starting program: /home/pabodamahagedara/Downloads/gdb_task/Executables/IT23623736
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at IT23623736.c:13
13      IT23623736.c: No such file or directory.
(gdb) 
```

```
(gdb) step
14      in IT23623736.c
(gdb) 
```

-Check output after each function

```
(gdb) info locals
fp = 0x0
uuid = '\000' <repeats 49 times>
key = 0x0
out = 0x0
(gdb) info args
No arguments.
```

```
(gdb) print key
$1 = 0x0
(gdb) █
```

- i. Determine the Contents of “data.txt”
 - a. Check if data.txt is encoded or encrypted

d.i. Explanation of its content

- ✓ Use file command to check file type

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ file data.txt
data.txt: very short file (no magic)
```

-Interpretation

- The file is extremely small
- It doesn't contain identifiable headers ("magic numbers") used by typical file types.
- It might contain raw or very minimal data possibility not even a full string.

- ✓ View file content in Human-readable format

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ cat data.txt
```

-Actual output is not visible, it may have

- Very short plain text
- Non-printable characters (which won't show up in terminal)
- A new line or whitespace only.

d.iii. Any decoded/processed information

- ✓ Try to decode as Base64

```
[pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ base64 -d data.txt  
base64: invalid input
```

-Interpretation

- The content is not valid Base64.
- The file likely has characters that don't conform to Base64 encoding.
- So it's not Base64-encoded.

- ✓ Try to convert from Hex (optional)

```
pabodamahagedara@pabodamahagedara-VirtualBox:~/Downloads/gdb_task/Executables$ xxd -r -p data.txt
```

- This usually implies the file is empty or has content not in hex format.
- This is not hex-encoded.

-If the file is mostly empty, contains only whitespace, or has a very short string, note that it is not encoded, encrypted, or compressed.

d.iv. Tools used for analysis

- file: Determines file type and encoding.
- cat/less: Views file content.
- strings: Extracts printable text.
- hexdump/xxd: Shows binary structure.
- base64: Checks for Base64 encoding.
- xxd -r: Tests for hex encoding.
- gdb: Debugs executable and traces file-writing functions.